PHYS4055

Atomic physics and quantum optics

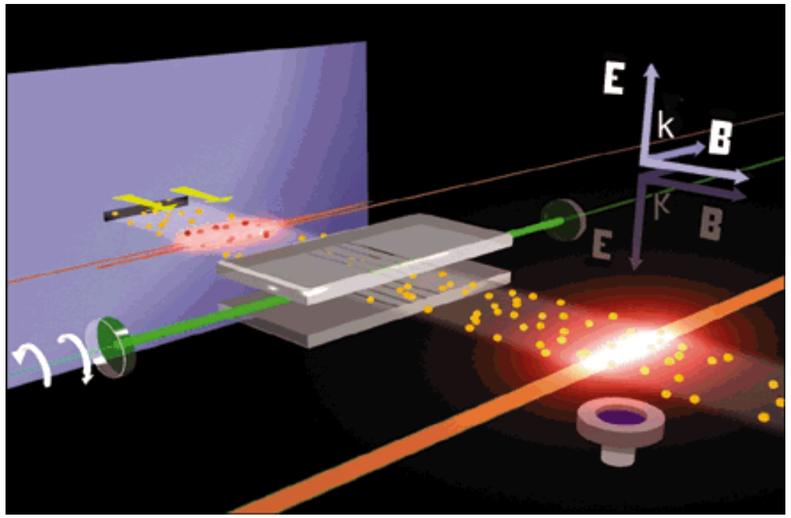
Week 12

Jacinda Ginges

j.ginges@uq.edu.au, room 330 Physics Annexe

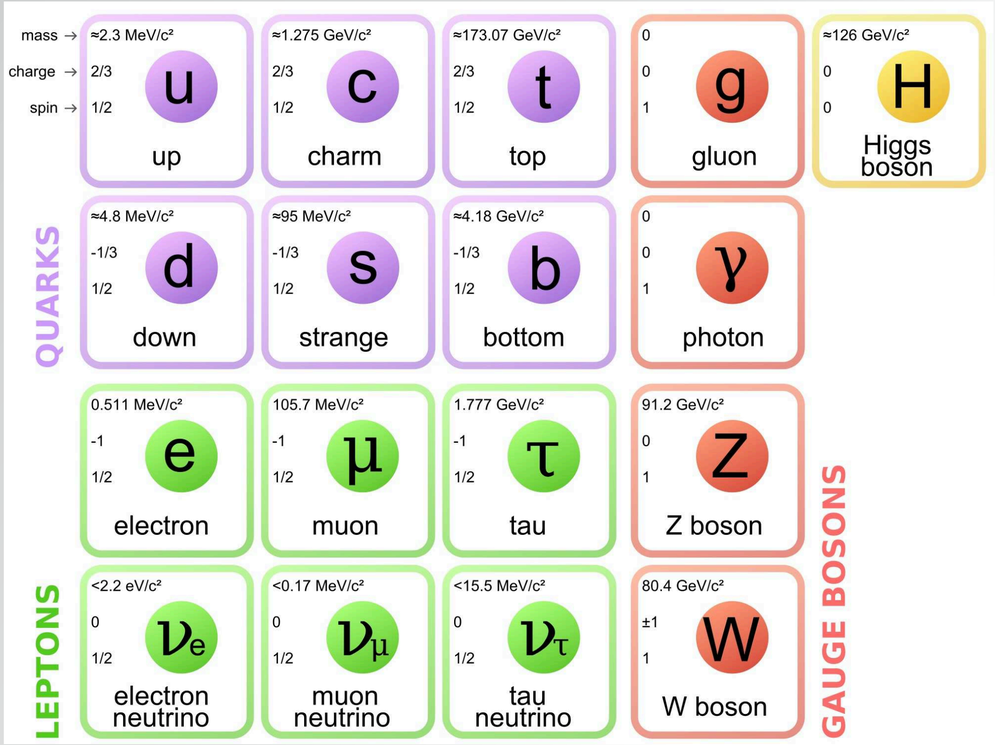
In these lectures, we will look into:

- fundamental symmetries violations in atoms:
  - atomic parity violation
  - electric dipole moments (CP violation)



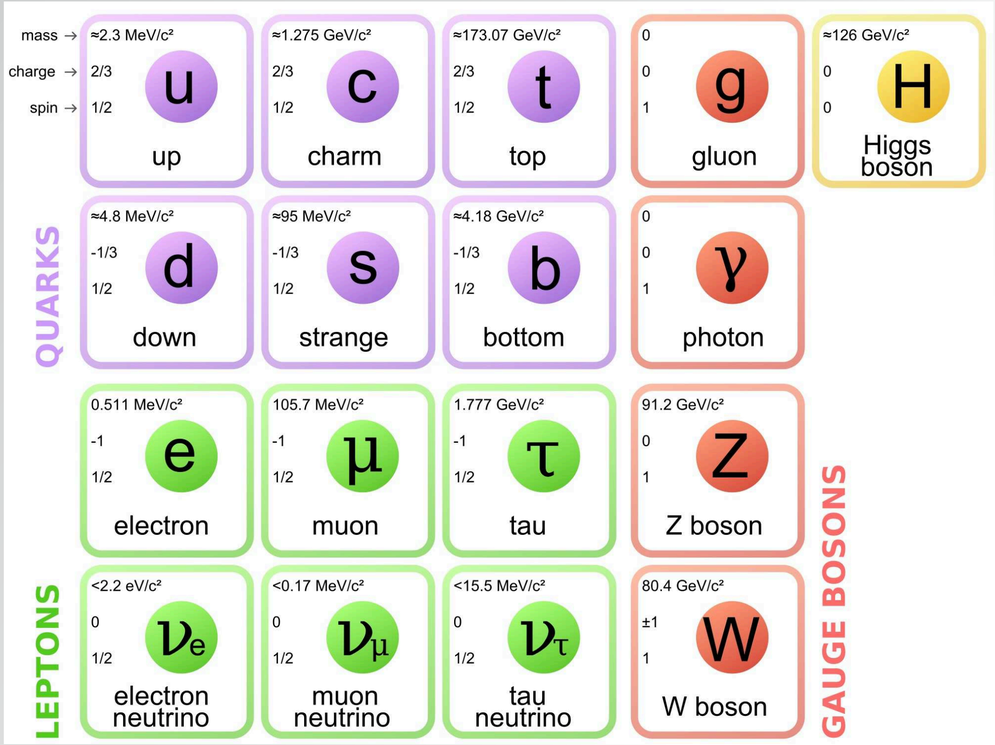
and tests of the Standard Model and searches for New Physics in atoms and molecules

Standard Model particle content



- All SM particles have been discovered!

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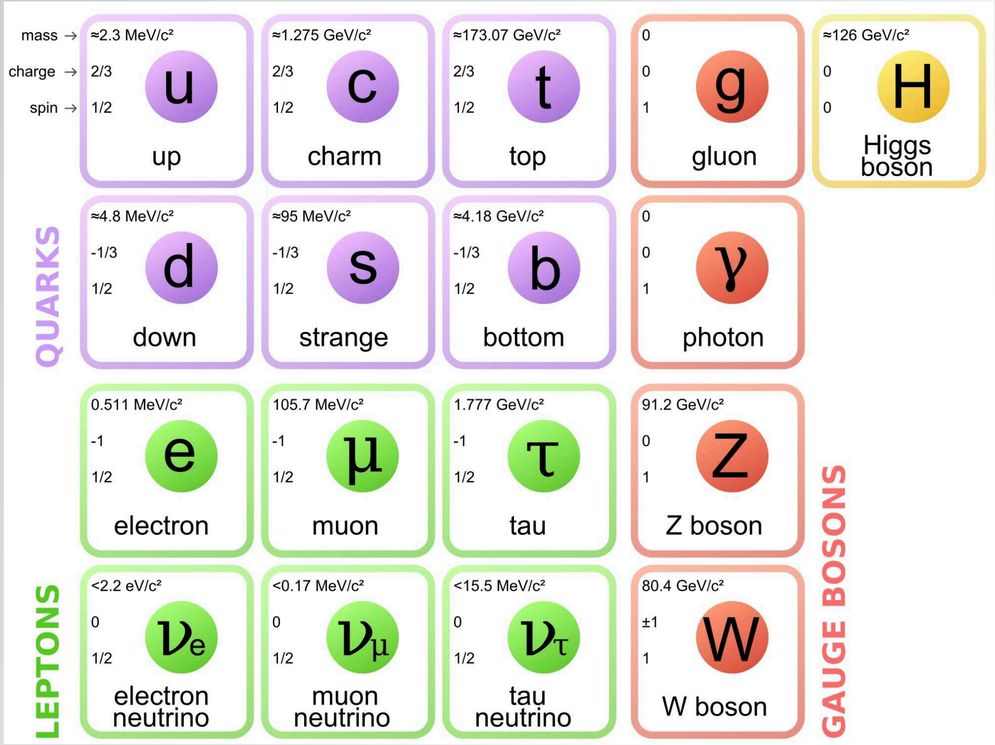
Searching for new physics

SM cannot explain

- Matter-antimatter asymmetry of universe



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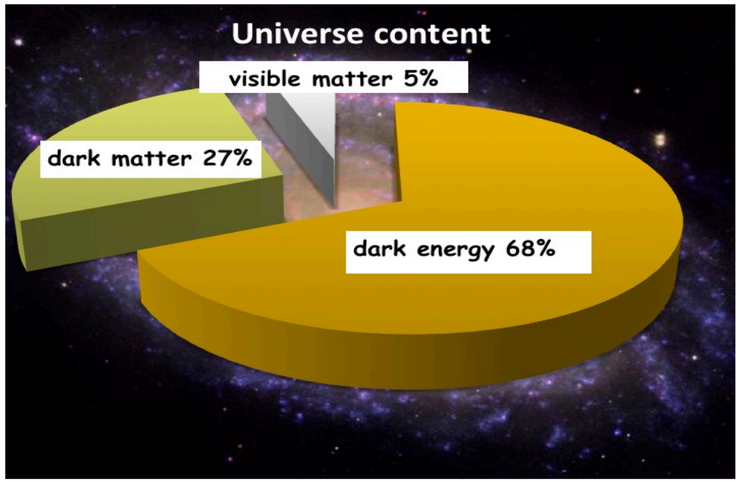
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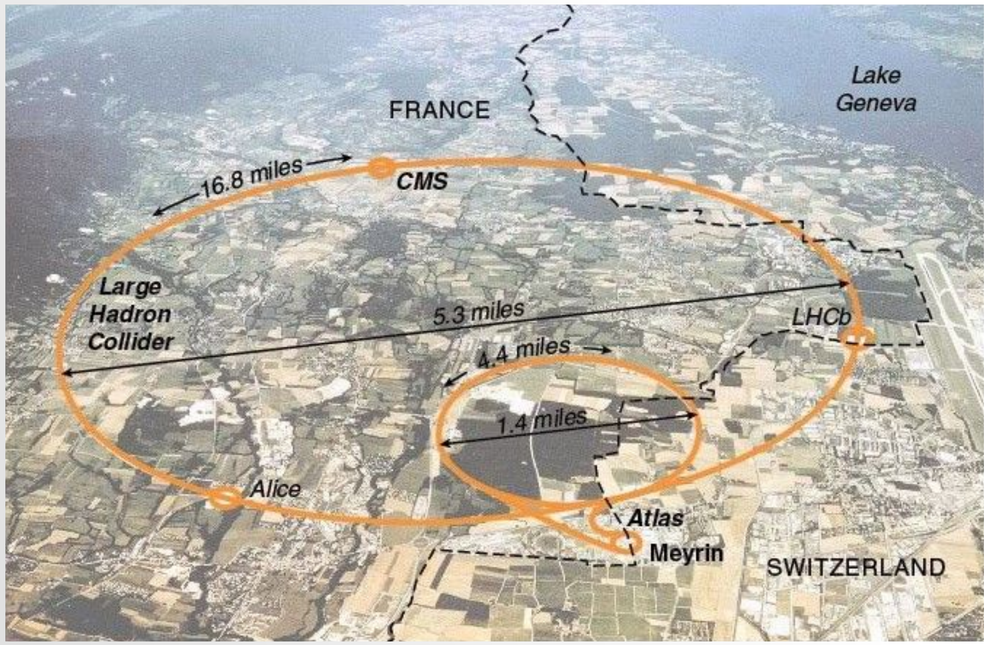
- Matter-antimatter asymmetry of universe





- Dark matter
- Dark energy

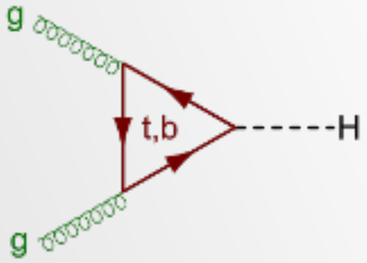
High energy



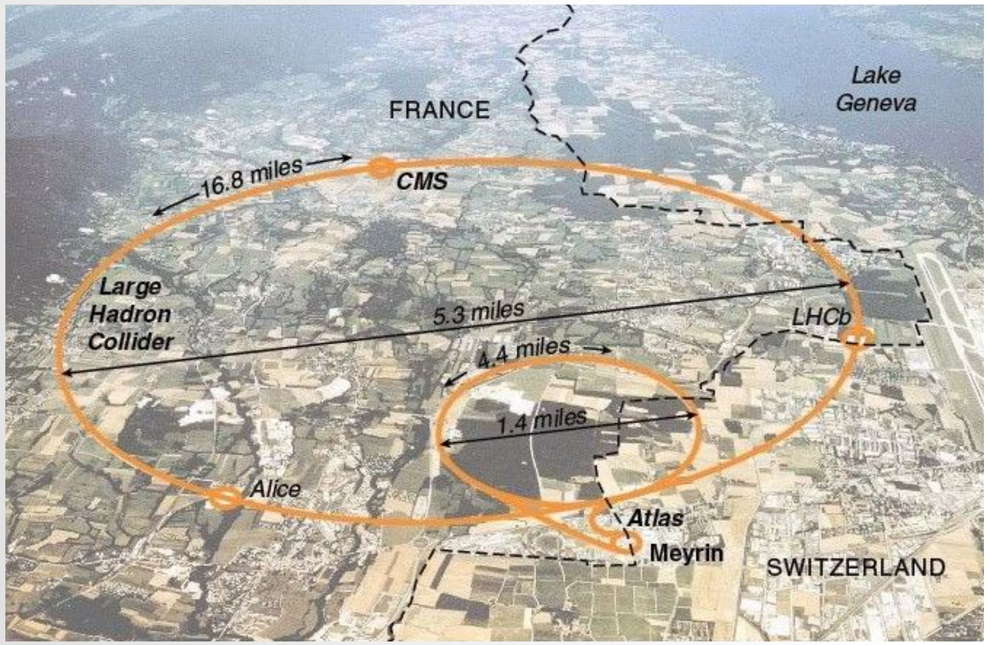
Large Hadron Collider, energies to 13 TeV

Produce particles *directly*

$$E = mc^2$$



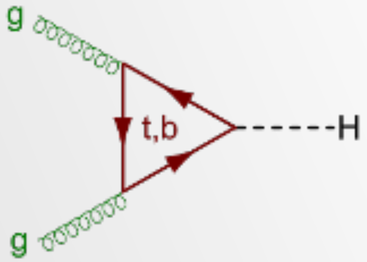
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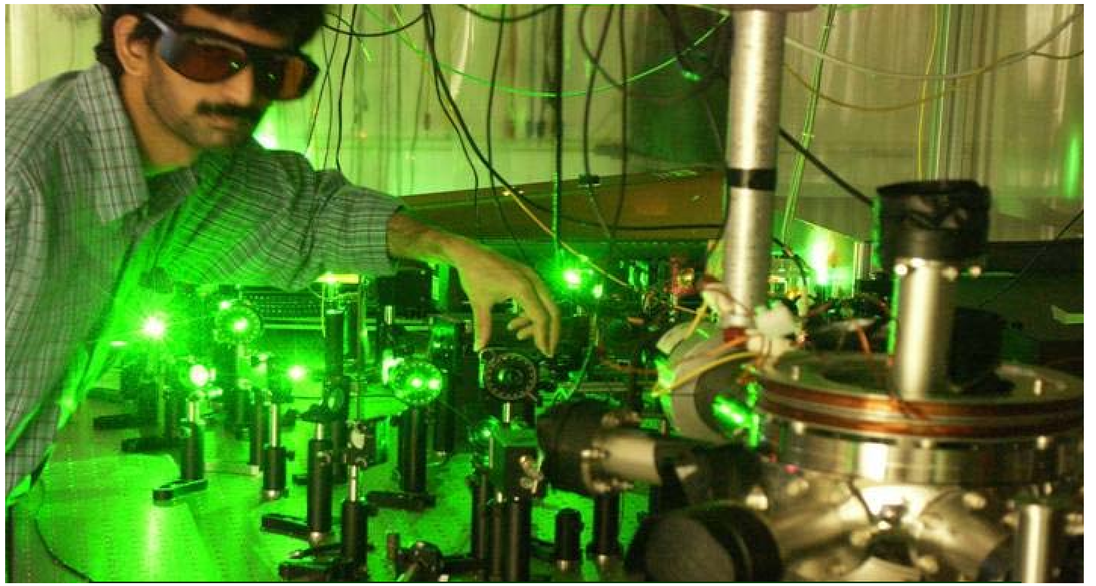
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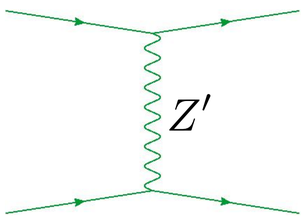
Low energy, high precision

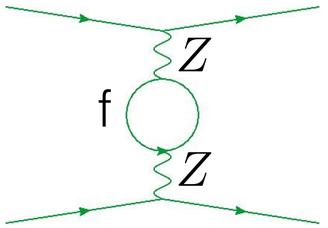


Probe *virtual* processes, may reach >> TeV



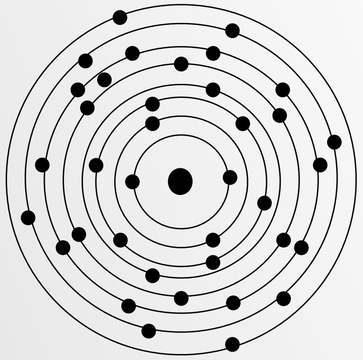
$$\Delta E \, \Delta t \sim \hbar$$





How can atoms be used to test the SM and search for new physics?

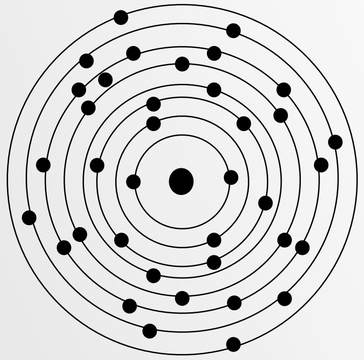
The atom as a laboratory for new physics searches



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- Weak interaction
- Strong interaction

are present in atoms and may be probed and tested

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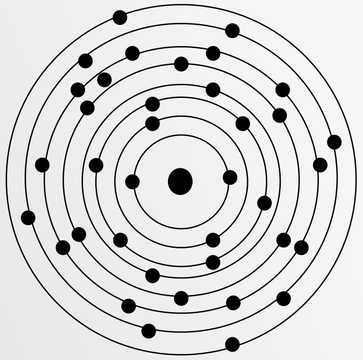
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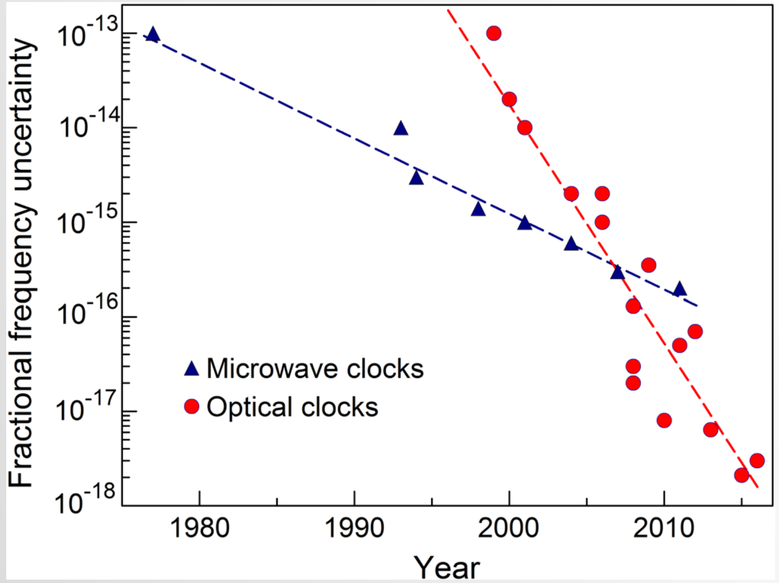
The complexity/simplicity of the system may be varied by changing nuclear charge (Z), isotope, ionisation degree, state

- Possibilities for enhancement
- May choose more theoretically tractable system

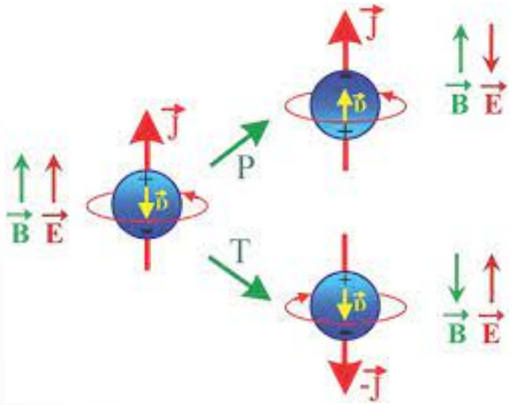
Searches for new physics in atomic experiments

Variation of fundamental constants

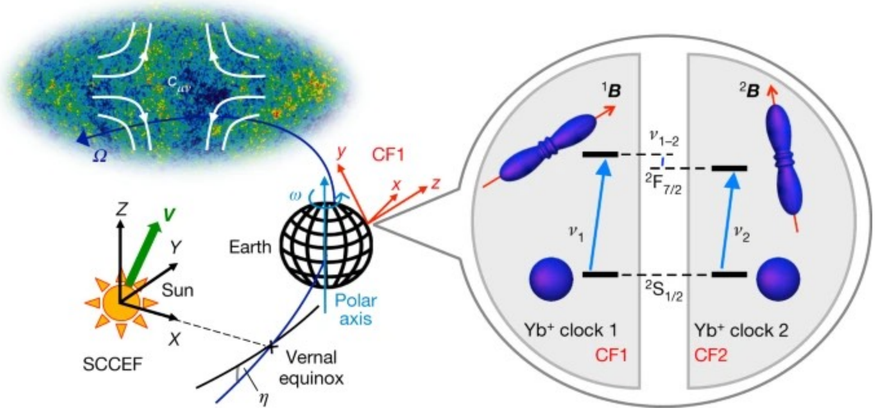
E.g., fine structure constant, proton-to-electron mass ratio, nuclear g-factors in clock experiments



Violation of fundamental symmetries P, CP, CPT



Tests of Lorentz symmetry, general relativity, gravitation,…

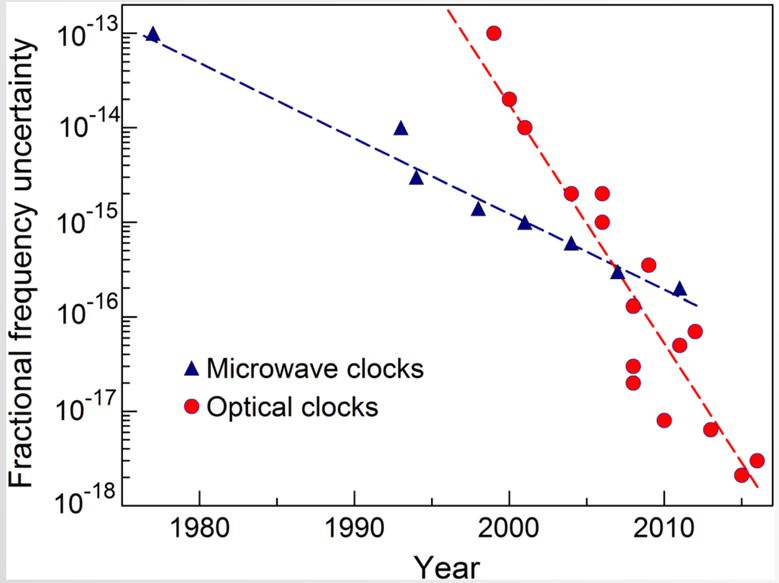


For a review, see Safronova et al., Rev. Mod. Phys. (2018)

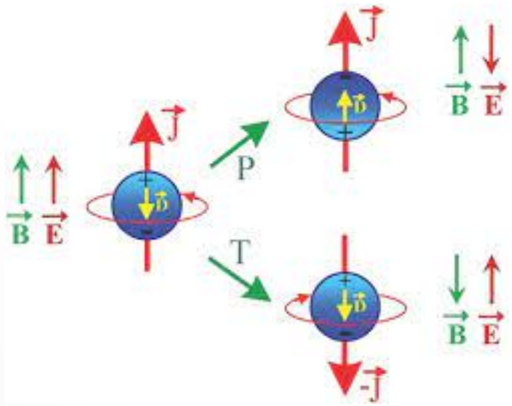
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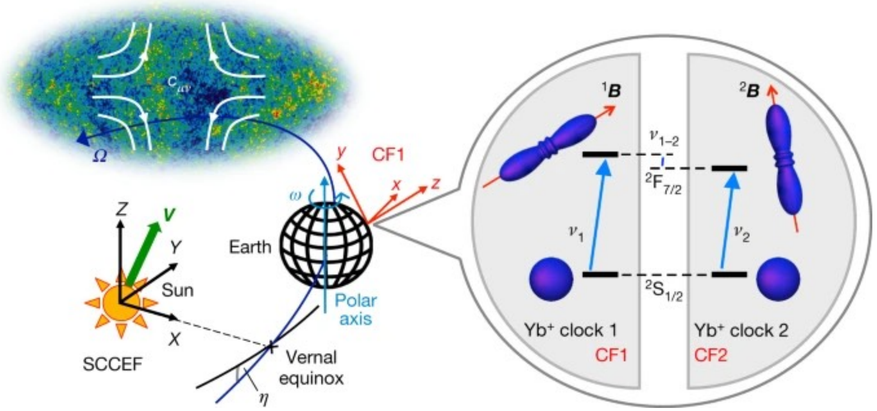
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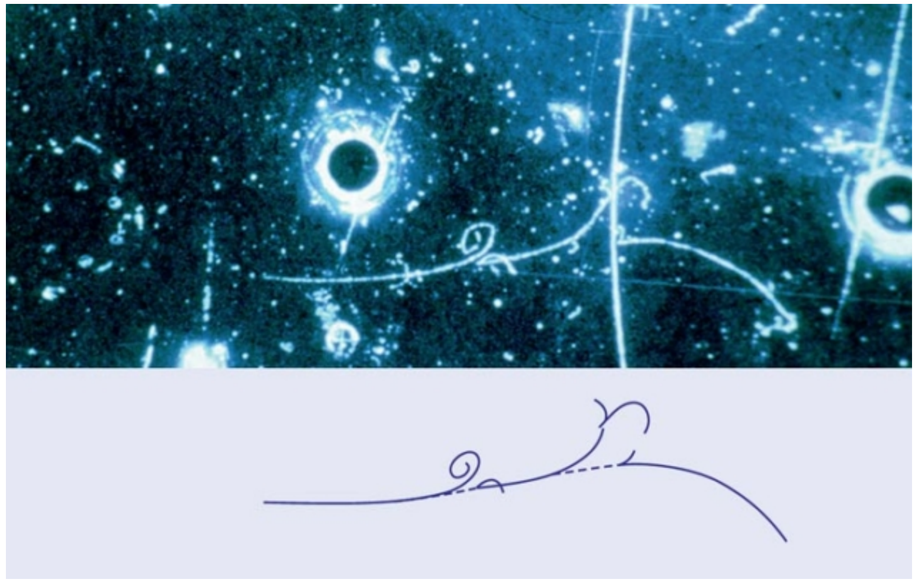
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Neutral weak currents

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Hadronic neutral current event: neutrino-nucleon scattering



Leptonic neutral current event: antineutrino-electron scattering

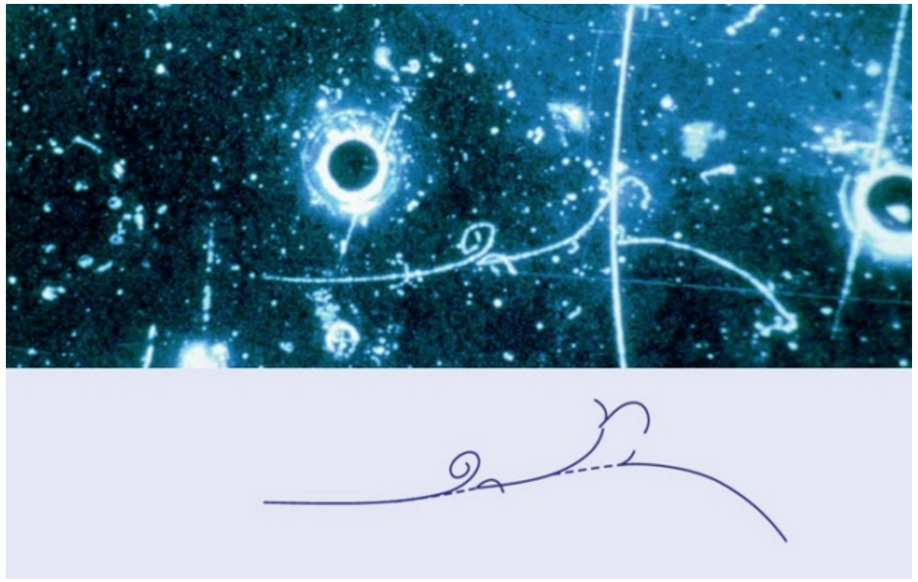
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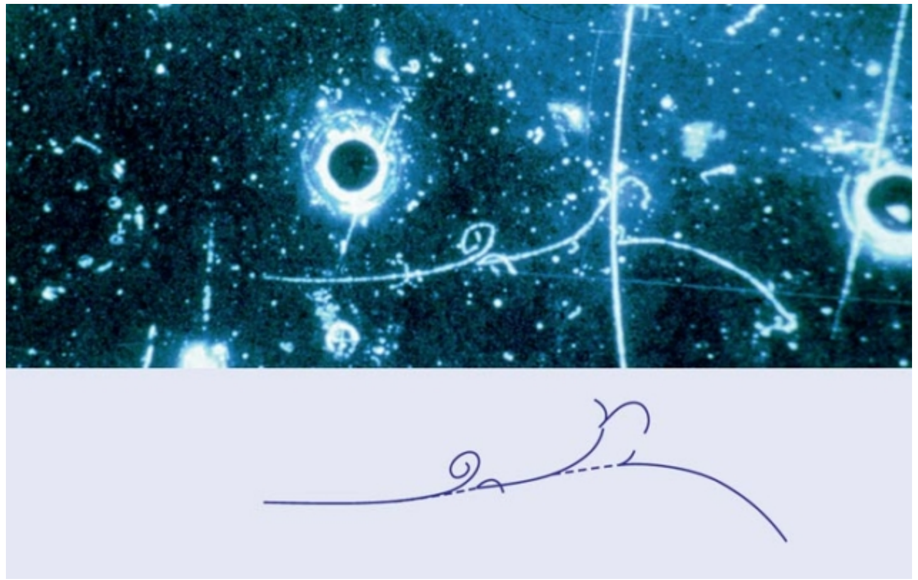
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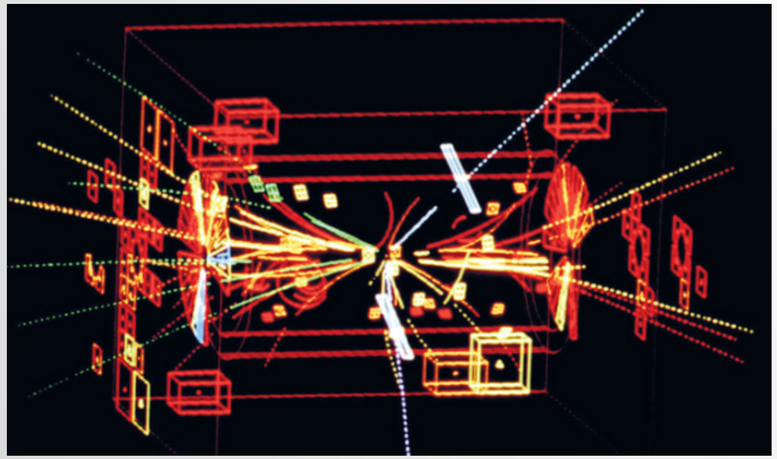


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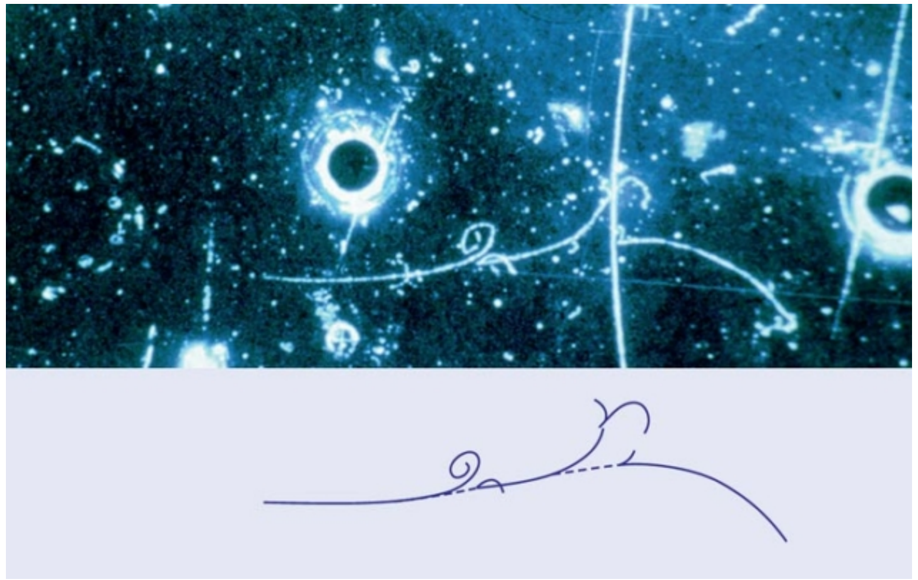


- In 1983, weak bosons Z, W^+, W^- produced directly at CERN

https://cerncourier.com/a/finding-the-w-and-z/



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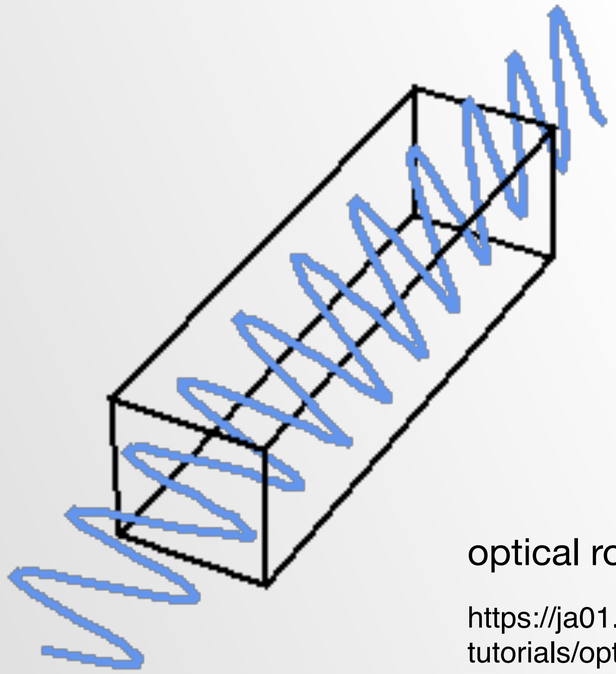


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Bismuth experiment

- e-N weak interaction produces optical activity
- Plane of polarisation of light is *rotated* on passing through bismuth vapour
- *Coherent, macroscopic parity-violating effect*

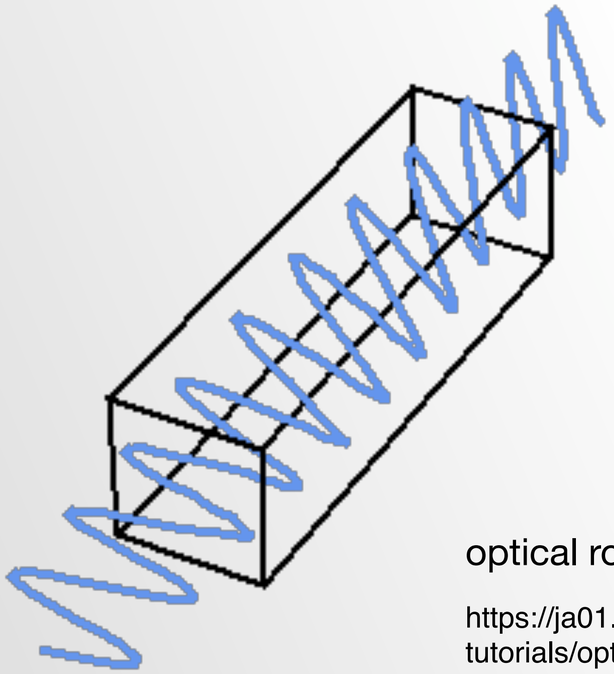


optical rotation animation

https://ja01.chem.buffalo.edu/tutorials/opticalactivity.html

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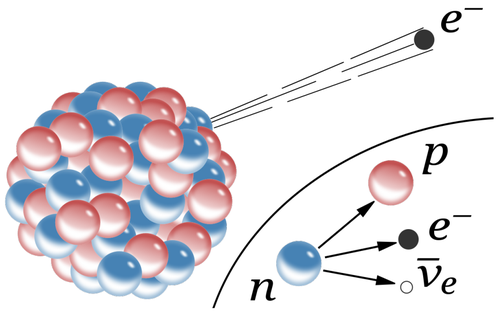
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The discovery of a new kind of a parity nonconserving weak interaction of electrons with nucleons is an example of a situation when a branch of physics (in this case, atomic spectroscopy) long since believed to be classical, again proves to be at the forefront of our understanding of nature… Table-top apparatus has proved to be an important addition to the experimental methods traditional for elementary particle physics. I am convinced that this case is not the last and that the time of table-top experiments in studying fundamental properties of matter is far from over.

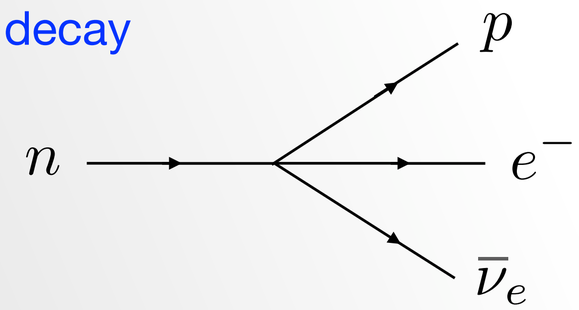
— *I. B. Khriplovich*

Fermi's four-fermion interaction

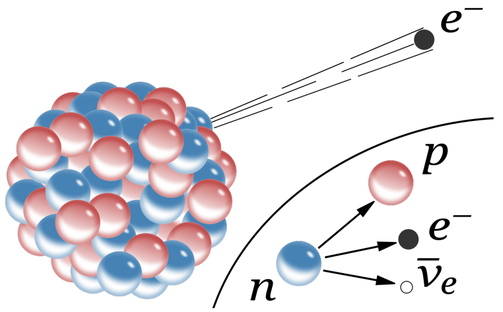


- Fermi (1934) constructed the first theory of the weak interaction
- Considered interaction to happen at a single point

beta decay

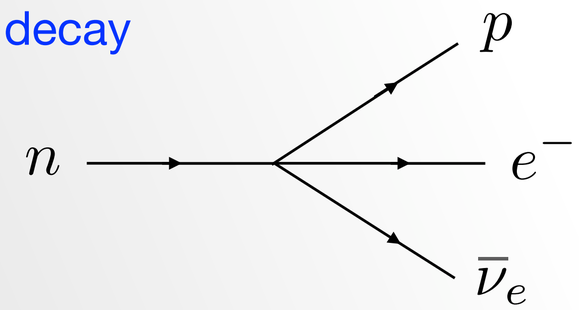


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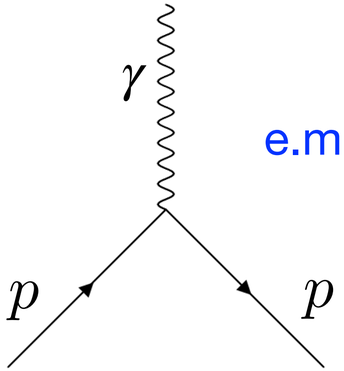
beta decay



$$\bar{p}\gamma_\mu n\,\bar{e}\gamma^\mu\nu_e$$

product of charged nucleon and lepton currents

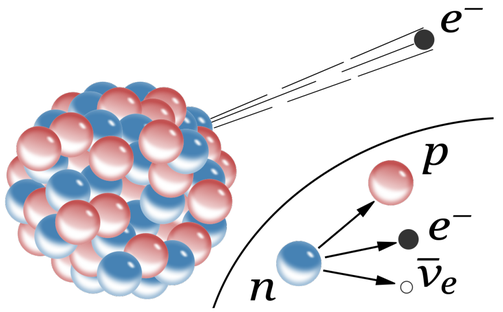
c.f.



e.m. interaction with proton

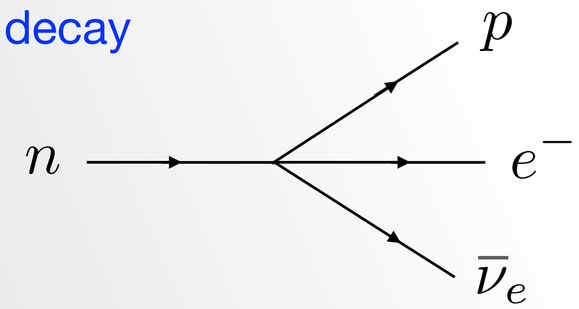
$$j_\mu A^\mu = \bar{p}\gamma_\mu p\,A^\mu$$

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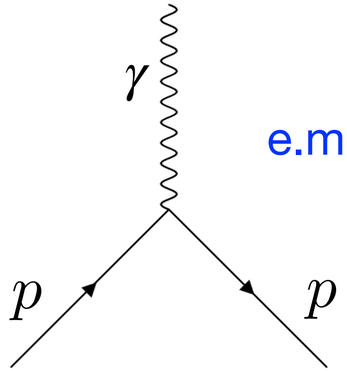
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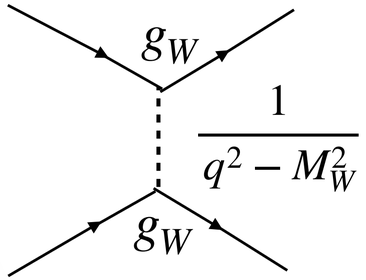
e.m. interaction with proton

$$j_\mu A^\mu = \bar{p}\gamma_\mu p\,A^\mu$$

- In the general case, can also construct (scalar) interaction from products of scalar (e.g., $\bar{p}n$), pseudoscalar ($\bar{p}\gamma_5 n$), axial-vector ($\bar{p}\gamma_\mu\gamma_5 n$), and antisymmetric tensor ($\bar{p}\sigma_{\mu\nu}n$) currents

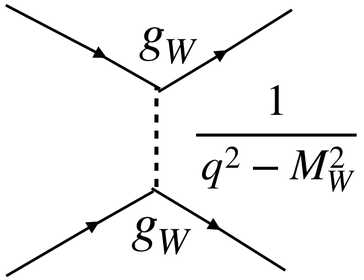
Consider exchange of W boson:

$$\text{Amplitude} \propto \frac{g_W^2}{q^2 - M_W^2}$$



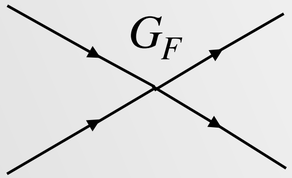
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- At low energies, $q^2 \ll M_W^2$, propagator $\frac{1}{q^2 - M_W^2} \to -\frac{1}{M_W^2}$

  $\to$ appears as a point-like interaction

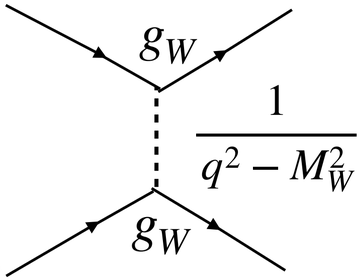


Fermi weak interaction constant G_F :

$$\frac{G_F}{\sqrt{2}} = \frac{g_W^2}{8M_W^2}$$

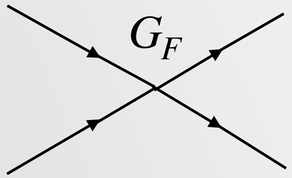
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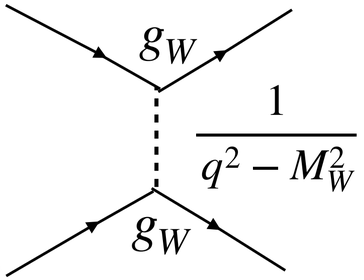
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- Weak bosons are very heavy, $W^{\pm}, Z^0 \approx 80 - 90\,\mathrm{GeV}/c^2$, so interaction range is very short, $\hbar/Mc \sim 10^{-3}\,\mathrm{fm}$

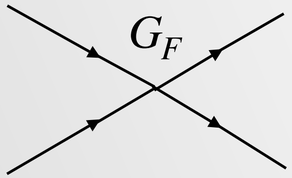
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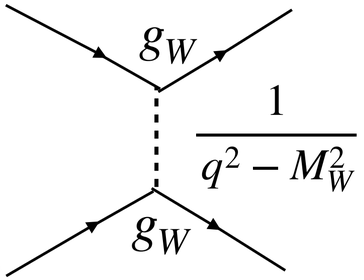
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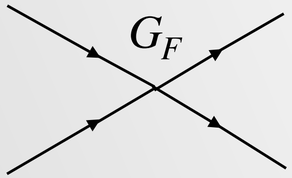
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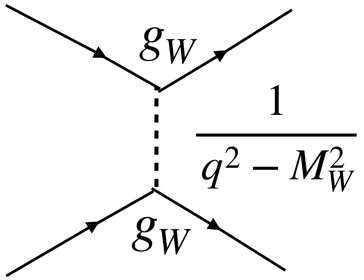
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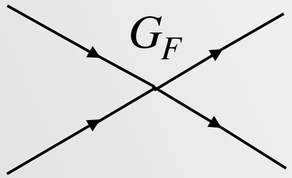
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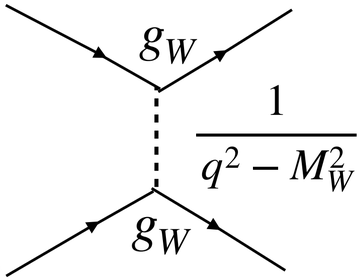
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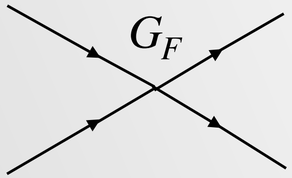
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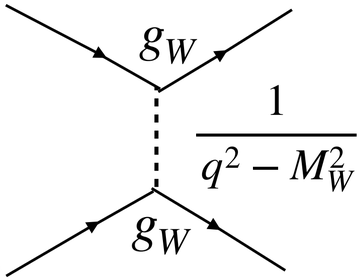
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P-odd, T-even — Atomic parity violation

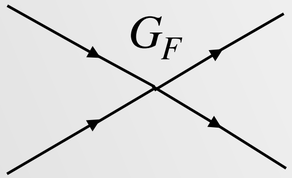
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Fundamental symmetries

The weak interaction violates fundamental symmetries *parity P* and the combined symmetry *charge conjugation and parity (CP).*

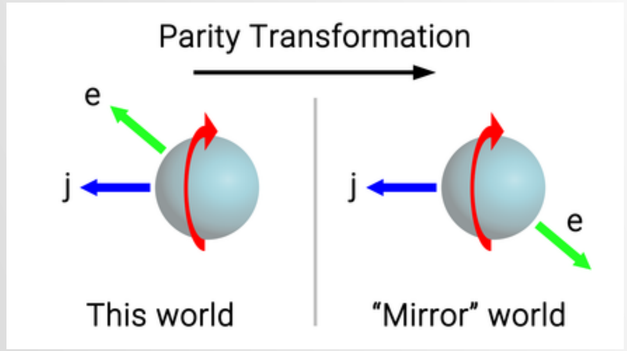
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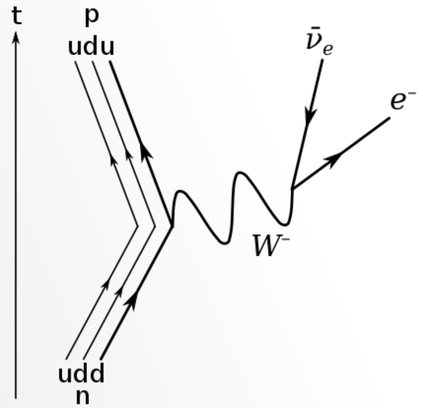
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Parity violation

First observed in 1957 in the beta decay of ^{60}Co.

$$^{60}_{27}\mathrm{Co} \rightarrow ^{60}_{28}\mathrm{Ni} + e^- + \bar{\nu}_e + 2\gamma$$





Foundational to SM.

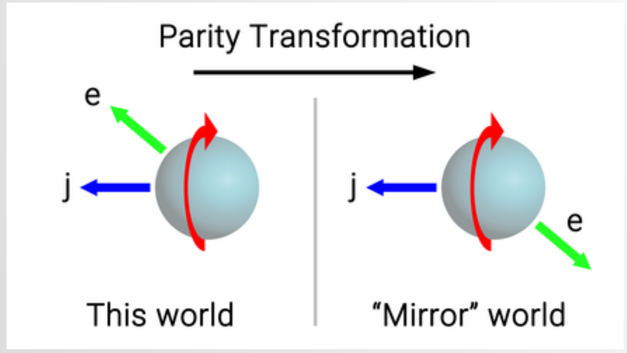
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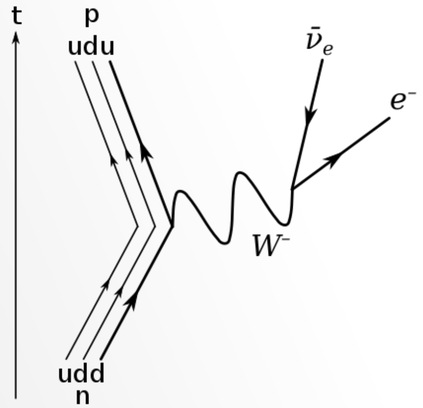
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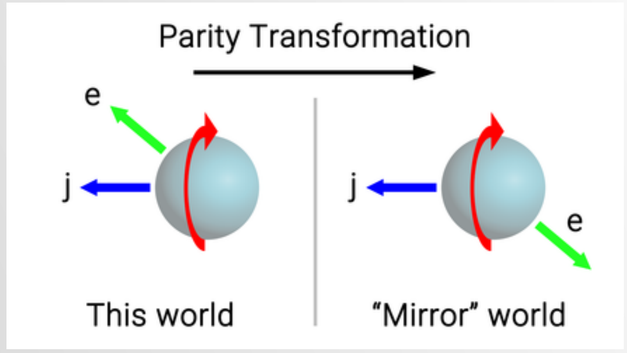
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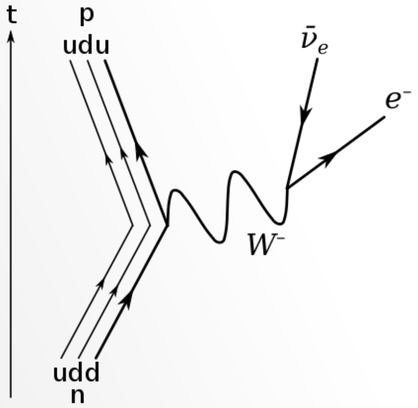
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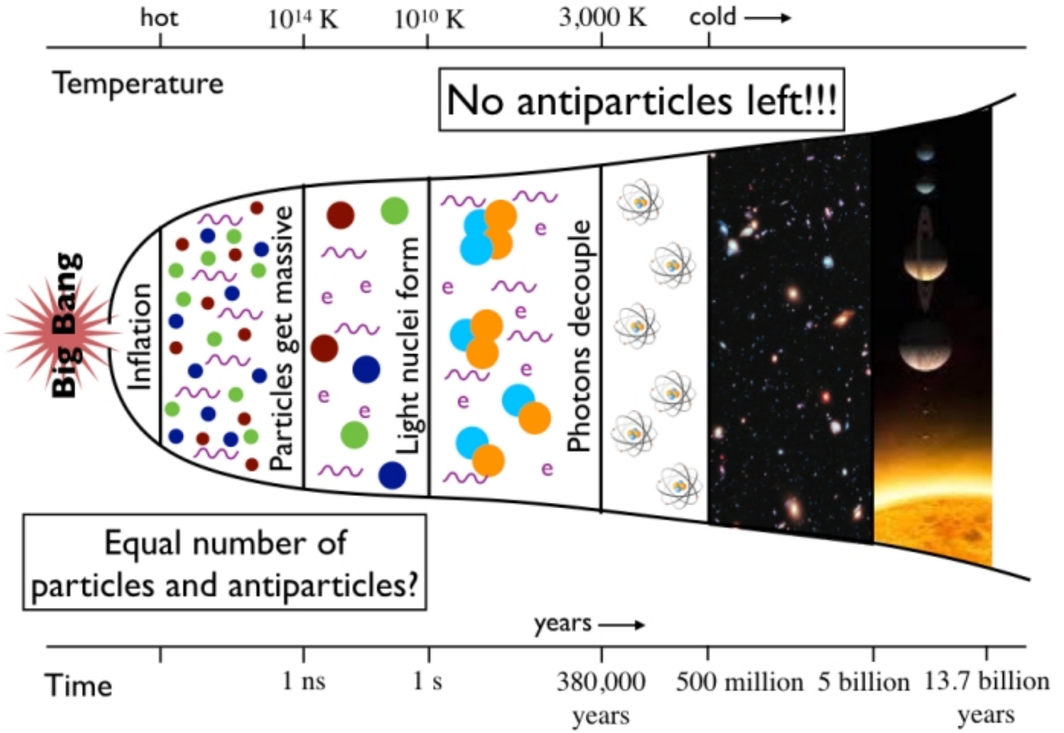


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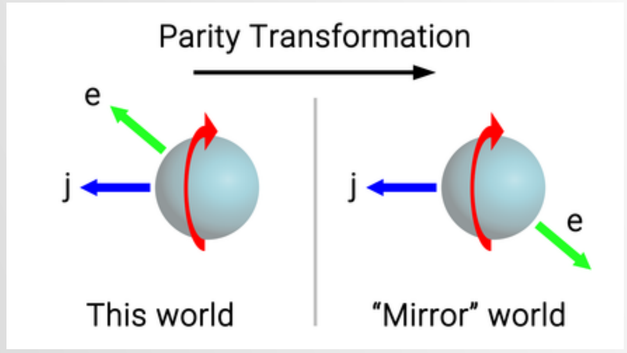
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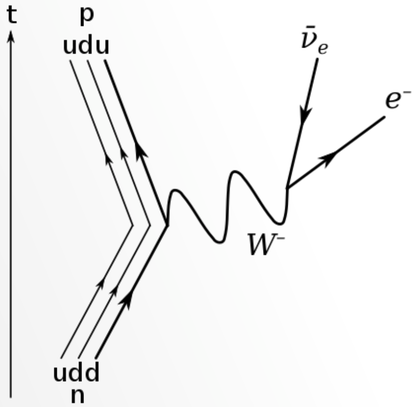
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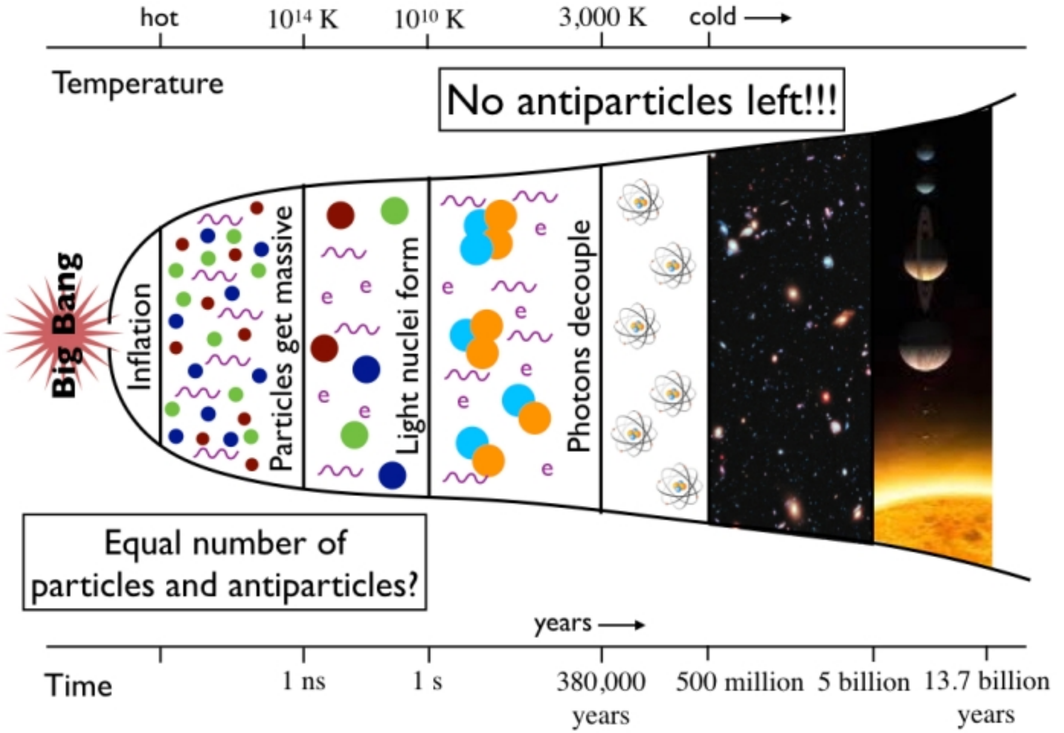


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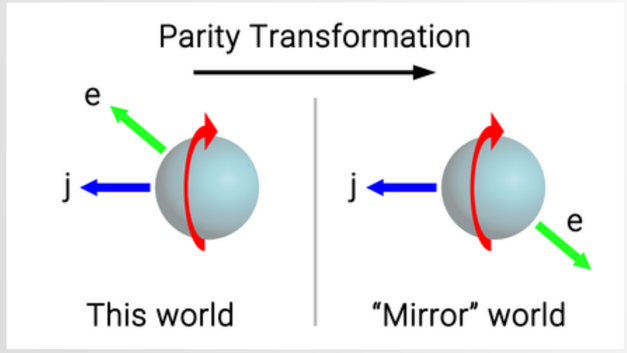
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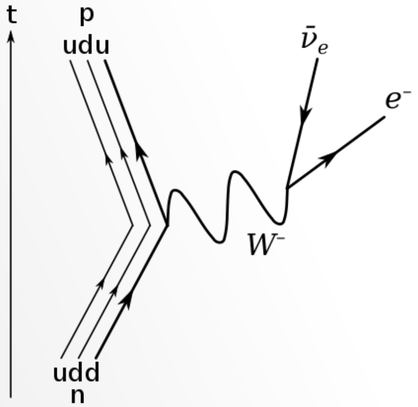
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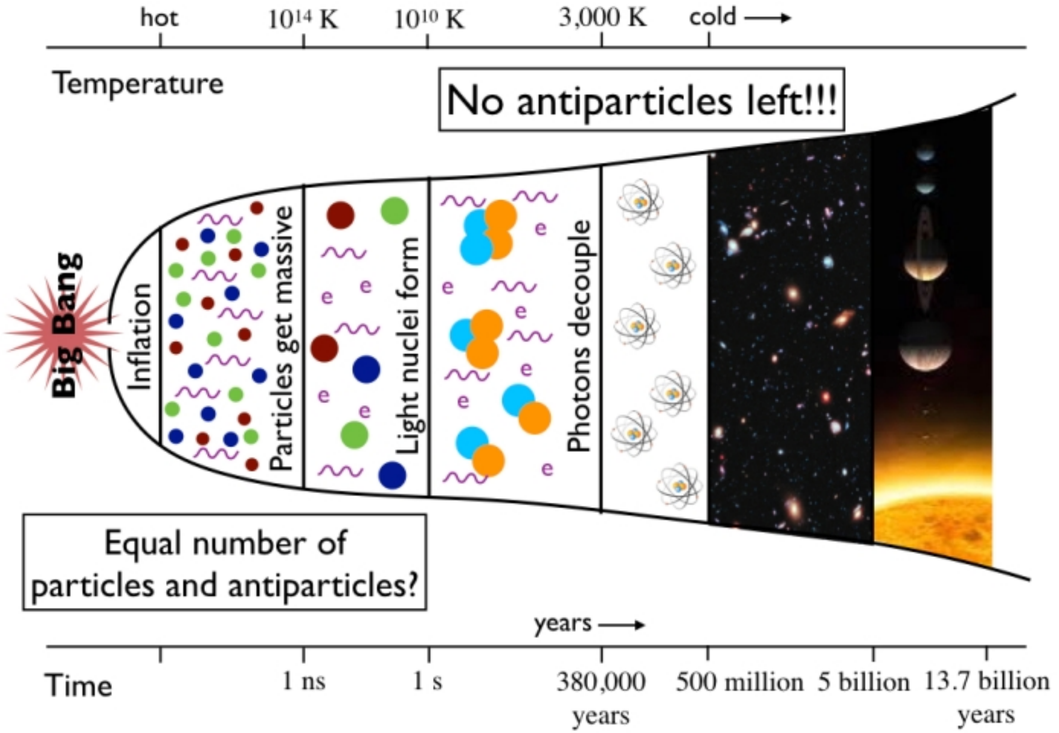


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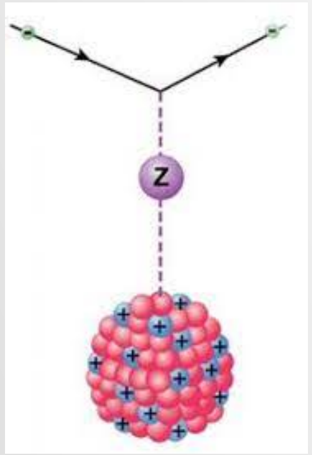
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Violations of fundamental symmetries in atoms

Precision atomic theory *needed* to extract fundamental parameters from atomic experiments for comparison with SM

Atomic parity violation (APV)



APV amplitude:

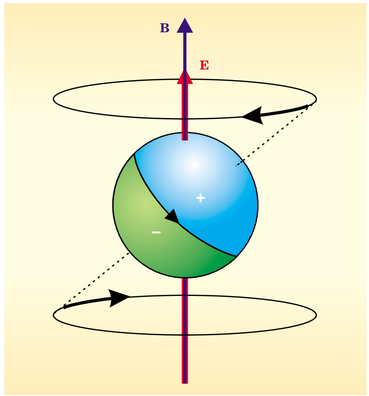
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from atomic structure theory

nuclear weak charge

Electric dipole moments (EDMs)

Parity- and time-reversal-violating



Atomic EDM:

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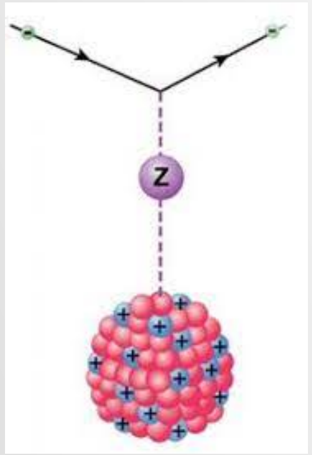
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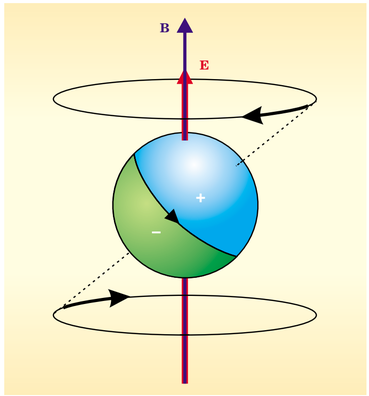
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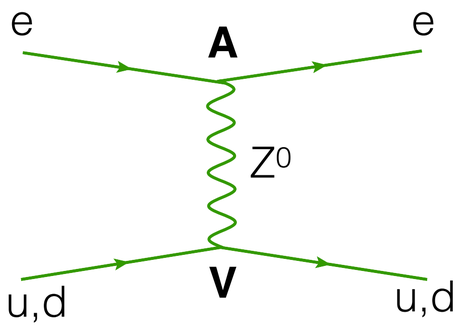
electron EDM

Atomic parity violation

Atomic parity violation and the nuclear weak charge

Axial vector coupling to electrons,
vector coupling to quarks

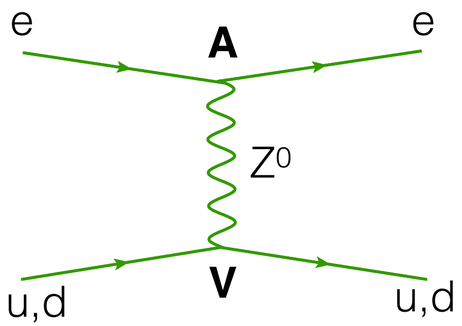
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Atomic parity violation and the nuclear weak charge

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Standard model tree-level couplings

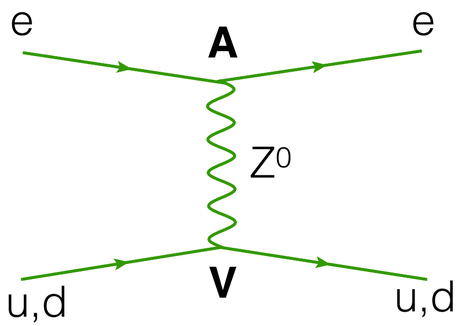
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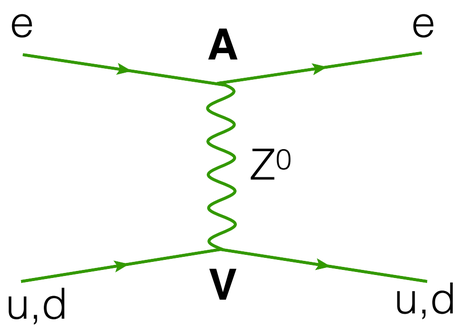
where Q_W is *nuclear weak charge.*

SM value known well, $Q_W^{\mathrm{SM}} = -73.23(1)$

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Parity-violating nature

Non-relativistic limit:

$$h_{\mathrm{PV}} \propto \boldsymbol{\sigma}\cdot\mathbf{p}$$

Parity operation:

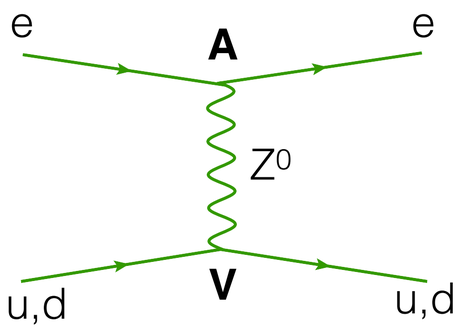
$$\boldsymbol{\sigma} \to \boldsymbol{\sigma}$$

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vector coupling to quarks

$$\frac{G}{\sqrt{2}} C_{1q} \left(\bar{e}\gamma_\mu\gamma_5 e\right)\left(\bar{q}\gamma^\mu q\right)$$



Standard model tree-level couplings

$$C_{1n} = C_{1u} + 2C_{1d} = -\frac{1}{2}$$

$$C_{1p} = 2C_{1u} + C_{1d} = \frac{1}{2}\left(1 - 4\sin^2\theta_W\right)$$

Leads to parity-violating interaction Hamiltonian for electrons

$$h_{\mathrm{PV}} = -\frac{G}{2\sqrt{2}} Q_W \rho(r)\gamma_5$$

where Q_W is *nuclear weak charge.*

SM value known well, $Q_W^{\mathrm{SM}} = -73.23(1)$

Parity-violating nature

Non-relativistic limit:

$$h_{\mathrm{PV}} \propto \boldsymbol{\sigma}\cdot\mathbf{p}$$

Parity operation:

$$\boldsymbol{\sigma} \to \boldsymbol{\sigma}$$
$$\mathbf{p} \to -\mathbf{p}$$

Enhancement with Z

Parity-violating amplitude:

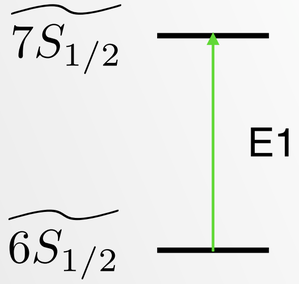
$$E_{\mathrm{PV}} \propto R(Z) Z^3$$

relativistic enhancement factor

nuclear charge

Bouchiat, Bouchiat (1974)

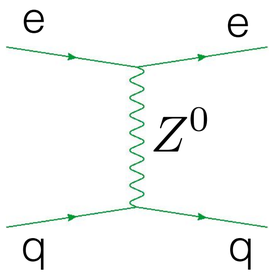
Atomic parity violation in cesium



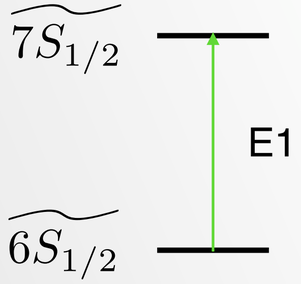
Weak interaction mixes opposite-parity states,

$$|\widetilde{S_{1/2}}\rangle = |S_{1/2}\rangle + \sum_n \frac{\langle nP_{1/2}|H_{\rm PV}|S_{1/2}\rangle}{E_{6S_{1/2}} - E_{nP_{1/2}}}|nP_{1/2}\rangle$$

6S - 7S electric dipole (E1) transition amplitude E_{PV}



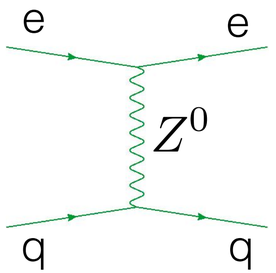
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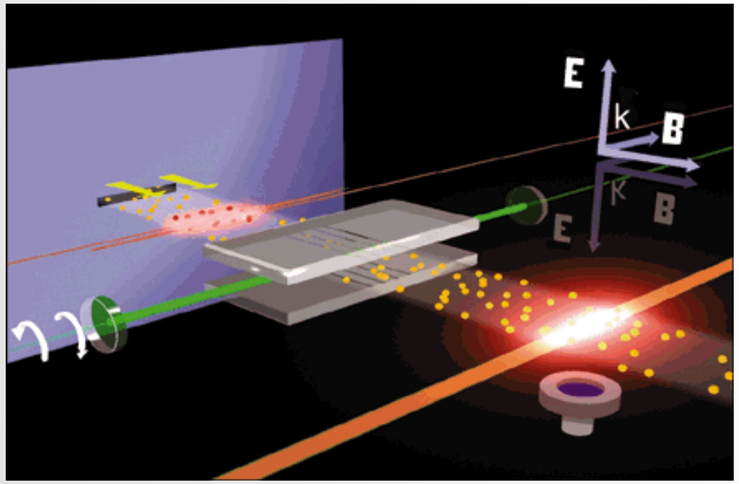
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6S - 7S electric dipole (E1) transition amplitude E_{PV}



Experiment, 0.35% uncertainty

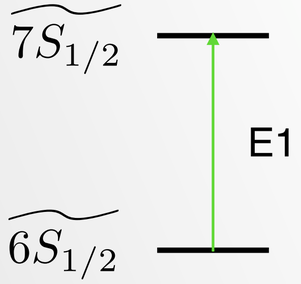


$$-\mathrm{Im}(E_{\rm PV})/\beta = 1.5935(1 \pm 0.35\%)\,\mathrm{mV/cm}$$

β — transition polarisability

Carl Wieman group, Wood et al., Science (1997)

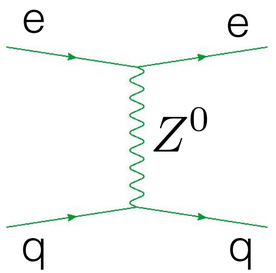
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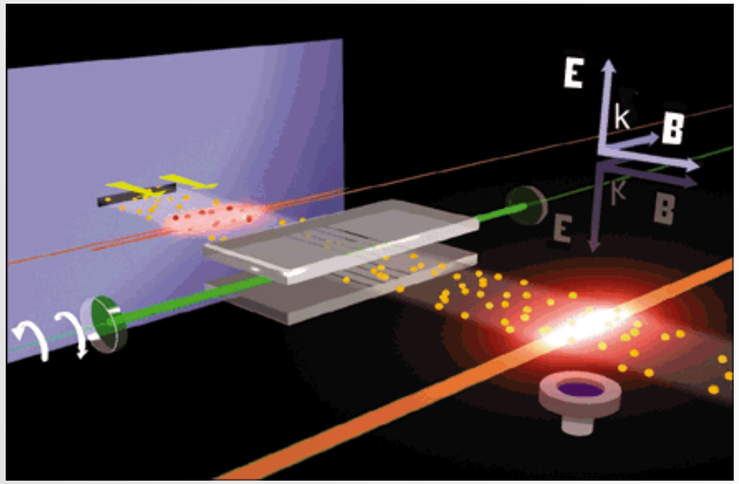
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Carl Wieman group, Wood et al., Science (1997)

Atomic theory, 0.5% uncertainty

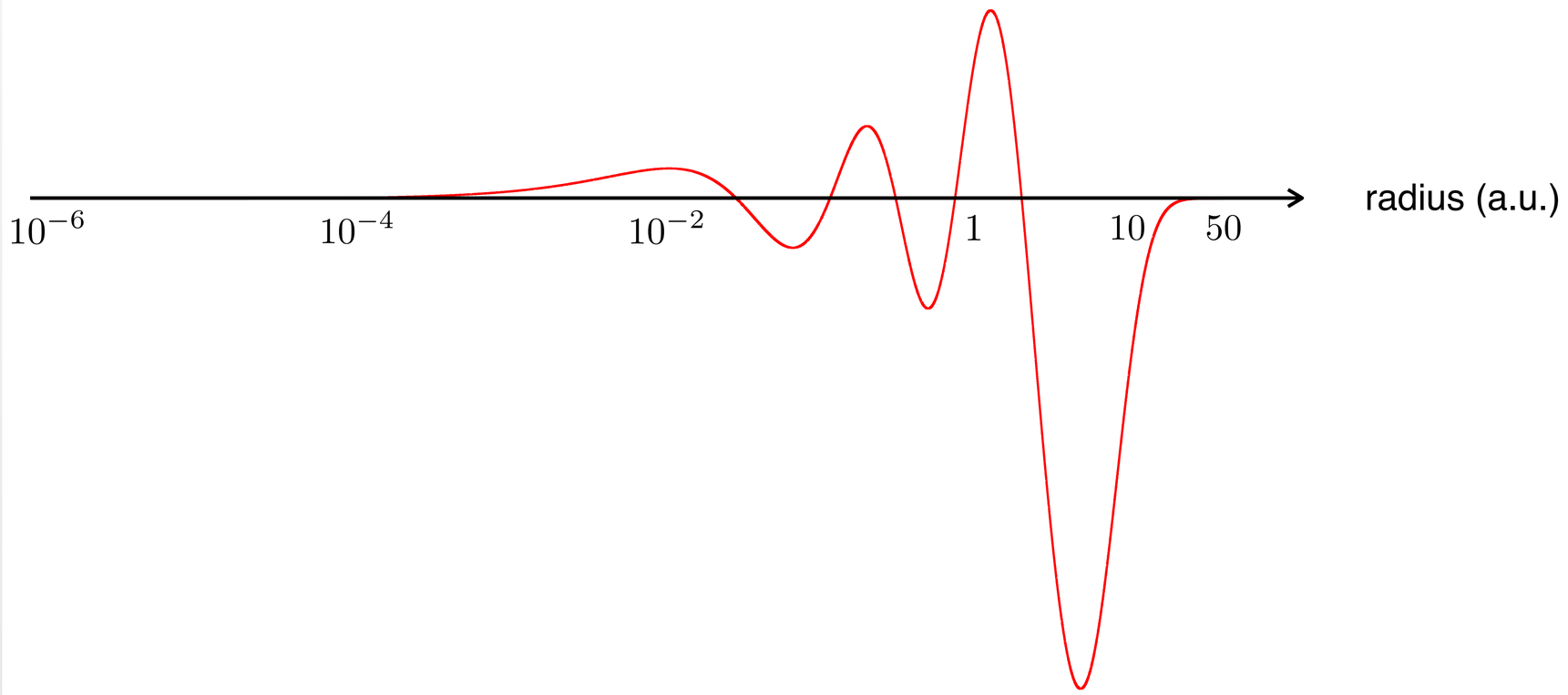
$$E_{\rm PV} = \langle\widetilde{7S_{1/2}}|D_z|\widetilde{6S_{1/2}}\rangle$$
$$= \sum_n \frac{\langle 7S_{1/2}|D_z|nP_{1/2}\rangle\langle nP_{1/2}|H_{\rm PV}|6S_{1/2}\rangle}{E_{6S_{1/2}} - E_{nP_{1/2}}}$$
$$= \xi\, Q_W$$

Dipole operator $\mathbf{D} = \sum_i e\mathbf{r}_i$, Weak operator $H_{\rm PV} = \sum_i (h_{\rm PV})_i$, Energies E

Dzuba, Flambaum, Ginges, PRD (2002); Flambaum, Ginges, PRA (2005)
Porsev, Beloy, Derevianko, PRL (2009); Dzuba, Berengut, Flambaum, Roberts, PRL (2012)

A note on theory and benchmarking

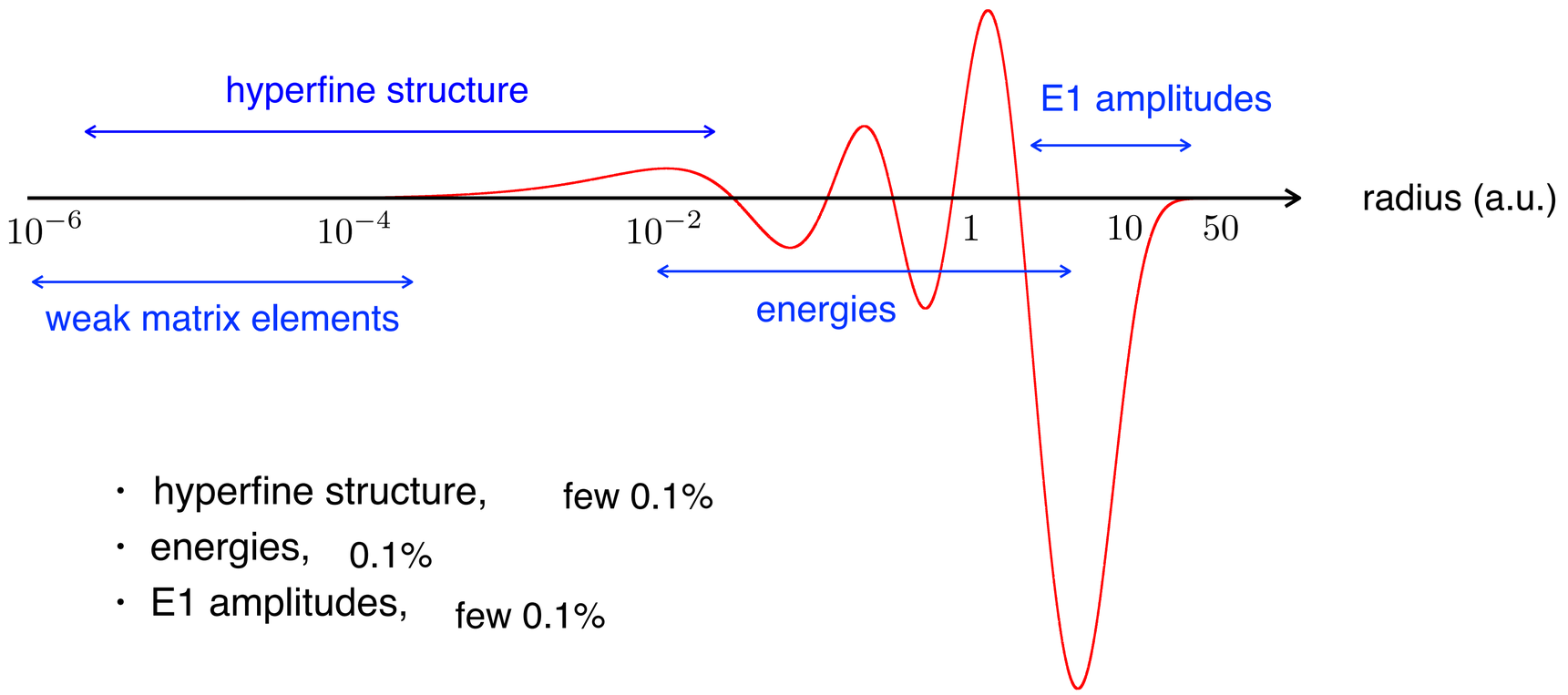
Upper radial component, Cs 6s:



$$E_{\mathrm{PV}} = \sum_n \frac{\langle 7S_{1/2}|D|nP_{1/2}\rangle\langle nP_{1/2}|H_{\mathrm{PV}}|6S_{1/2}\rangle}{E_{6S_{1/2}} - E_{nP_{1/2}}} + \sum_n \frac{\langle 7S_{1/2}|H_{\mathrm{PV}}|nP_{1/2}\rangle\langle nP_{1/2}|D|6S_{1/2}\rangle}{E_{7S_{1/2}} - E_{nP_{1/2}}} = \xi Q_W$$

A note on theory and benchmarking

Upper radial component, Cs 6s:



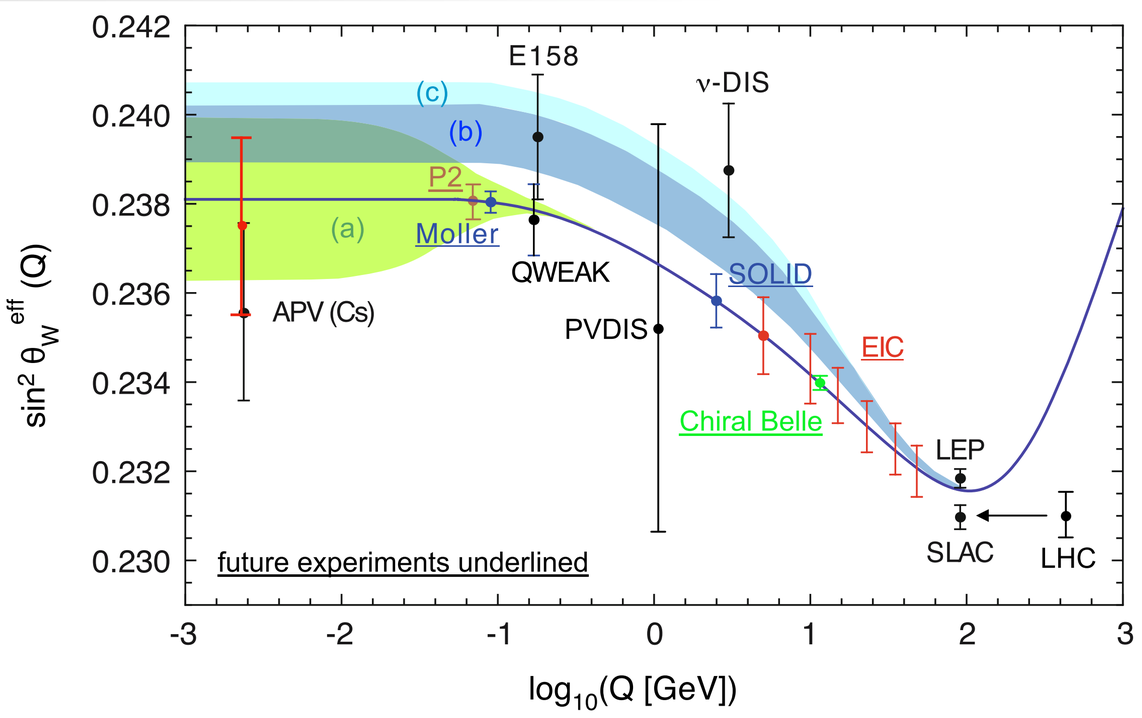
- hyperfine structure, few 0.1%
- energies, 0.1%
- E1 amplitudes, few 0.1%

$$E_{\rm PV} = \sum_n \frac{\langle 7S_{1/2}|D|nP_{1/2}\rangle\langle nP_{1/2}|H_{\rm PV}|6S_{1/2}\rangle}{E_{6S_{1/2}} - E_{nP_{1/2}}} + \sum_n \frac{\langle 7S_{1/2}|H_{\rm PV}|nP_{1/2}\rangle\langle nP_{1/2}|D|6S_{1/2}\rangle}{E_{7S_{1/2}} - E_{nP_{1/2}}} = \xi Q_W$$

Tests of the standard model

- Experiment and theory: nuclear weak charge: $Q_W = -73.07(28)(33) \quad \Rightarrow \quad Q_W - Q_W^{\mathrm{SM}} = 0.16(43)$

Running of the Weinberg angle



Weinberg angle — aka “weak mixing angle”

- fundamental parameter of SM
- Mixes W and B fields to produce the massive Z^0 and massless photon
- At tree level:

$$\sin^2\theta_W = 1 - \frac{m_W^2}{m_Z^2}$$

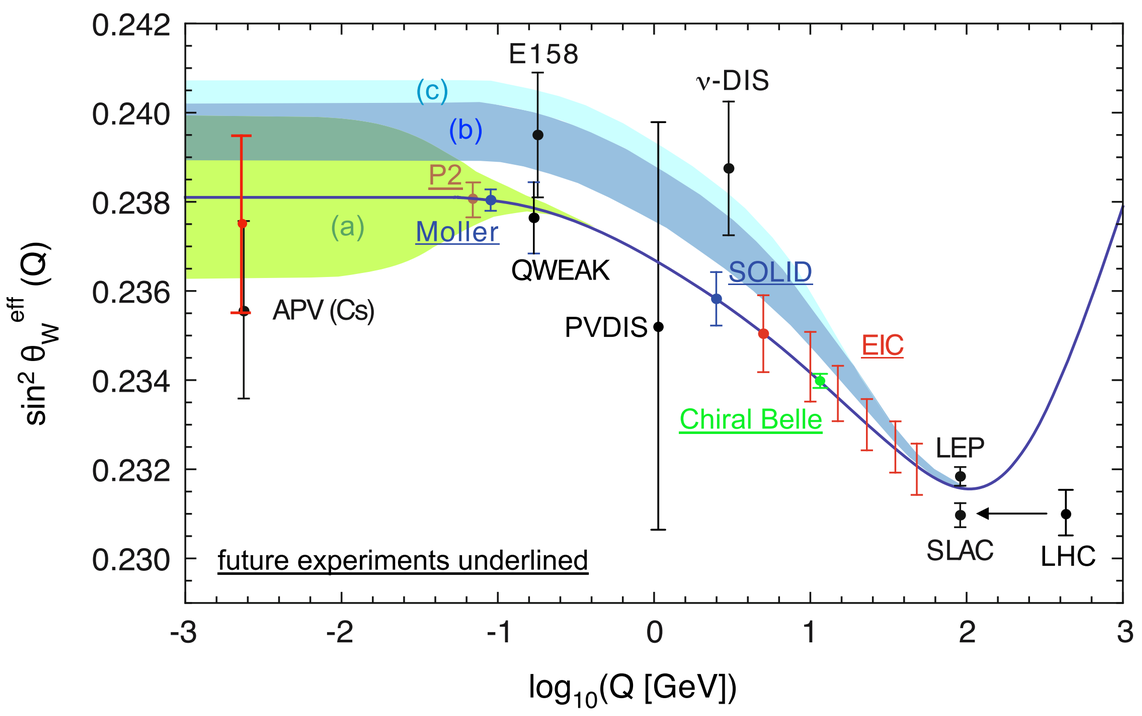
Figure from: Gwinner and Orozco, Quantum Sci. Technol (2022)

New result for vector polarizability shifts APV result (red): G. Toh et al., PRL (2019)

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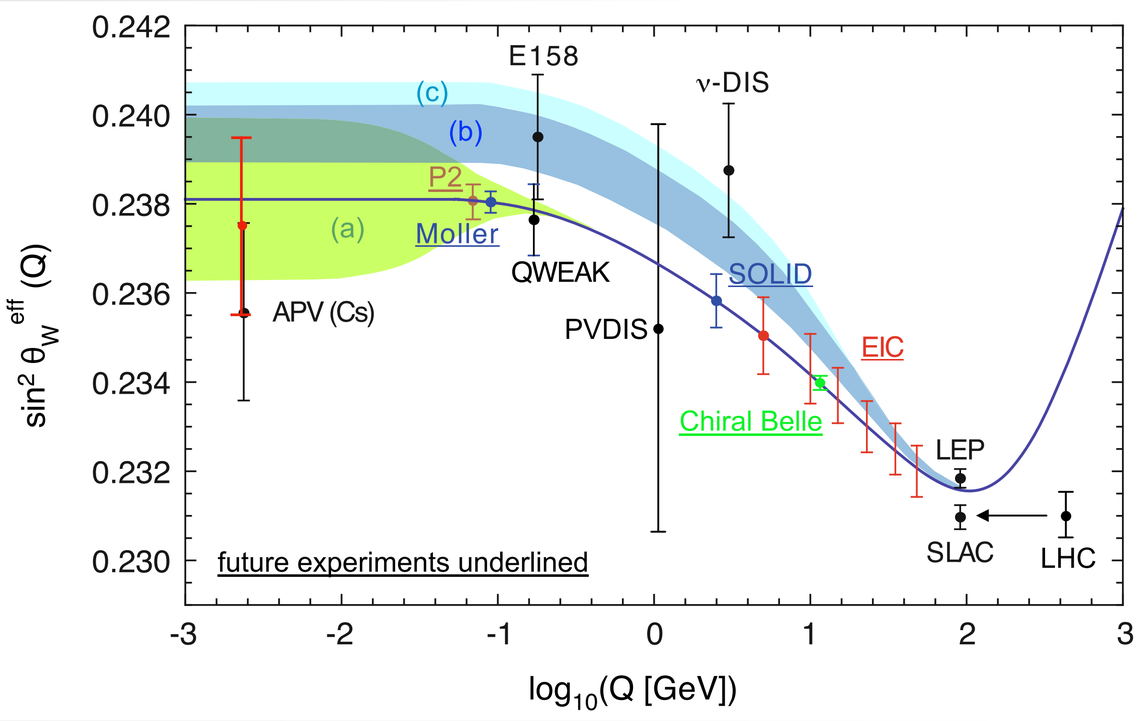
QWEAK - electron-proton scattering Ross Young
E158 - electron-electron scattering @ SLAC
PVDIS - parity-violation in deep inelastic scattering
ν-DIS - neutrino deep inelastic scattering
Tevatron - proton-antiproton collider
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Dark Z boson:
(a) 50 MeV; (b) 15 MeV; (c) 15 MeV, in tension with expt.

Figure from: Gwinner and Orozco, Quantum Sci. Technol (2022)
New result for vector polarizability shifts APV result (red): G. Toh et al., PRL (2019)

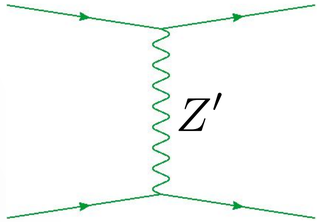
Searches for new physics

New tree-level physics

$$Q_W = Q_W^{\mathrm{SM}} + \Delta Q_W$$

$$\mathcal{L}_{\mathrm{PV}} = \mathcal{L}_{\mathrm{PV}}^{\mathrm{SM}} + \mathcal{L}_{\mathrm{PV}}^{\mathrm{new}}$$

e.g.

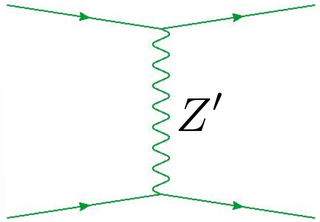


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$$\mathcal{L}_{\rm PV} = \mathcal{L}_{\rm PV}^{\rm SM} + \mathcal{L}_{\rm PV}^{\rm new}$$

$$\mathcal{L}_{\rm PV}^{\rm SM} = \frac{G_F}{2\sqrt{2}} g_A^e \bar{e}\gamma_\mu\gamma_5 e \sum_q g_V^q \bar{q}\gamma^\mu q$$

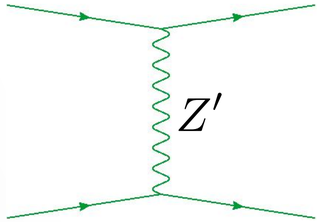
$$\mathcal{L}_{\rm PV}^{\rm new} = \frac{4\pi\kappa^2}{\Lambda^2} \bar{e}\gamma_\mu\gamma_5 e \sum_q h_V^q \bar{q}\gamma^\mu q$$

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Probing mass scale:

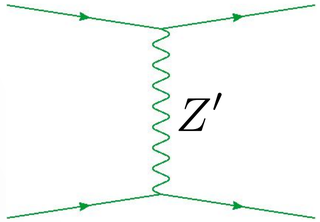
$$\Lambda \geq \left(\frac{8\sqrt{2}\pi\kappa^2}{(\Delta Q_W / Q_W^{\rm SM})\, G_F} \right)^{1/2}$$

Searches for new physics

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$$Q_W = Q_W^{\rm SM} + \Delta Q_W$$

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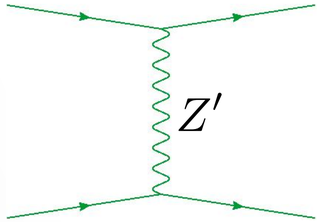
for 0.5%

Searches for new physics

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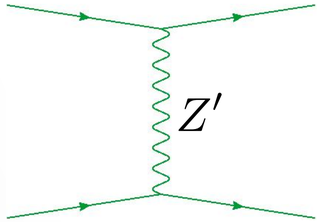
strongly interacting $\kappa^2 \sim 1$, weakly interacting $\kappa^2 \sim \alpha$

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Z' boson: $m_{Z'} \gtrsim 1\,\text{TeV}$

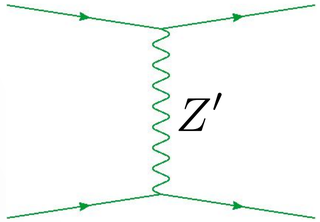
Ramsey-Musolf, PRC (1999)

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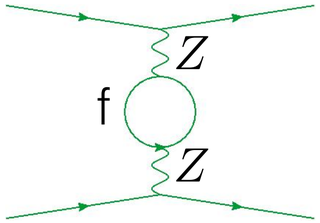
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strongly interacting $\kappa^2 \sim 1$, weakly interacting $\kappa^2 \sim \alpha$

Z' boson: $m_{Z'} \gtrsim 1\ {\rm TeV}$

Ramsey-Musolf, PRC (1999)

New radiative corrections



Defined in terms of Peskin-Takeuchi parameters S, T

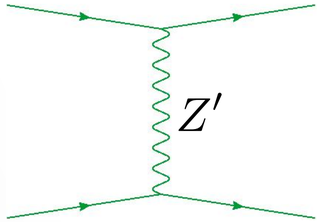
$$Q_W - Q_W^{\rm SM} = -0.800\, S - 0.007\, T$$

Searches for new physics

New tree-level physics

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e.g.



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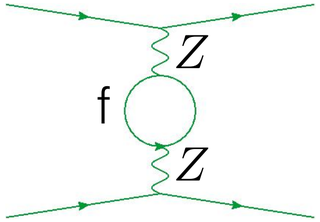
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Ramsey-Musolf, PRC (1999)

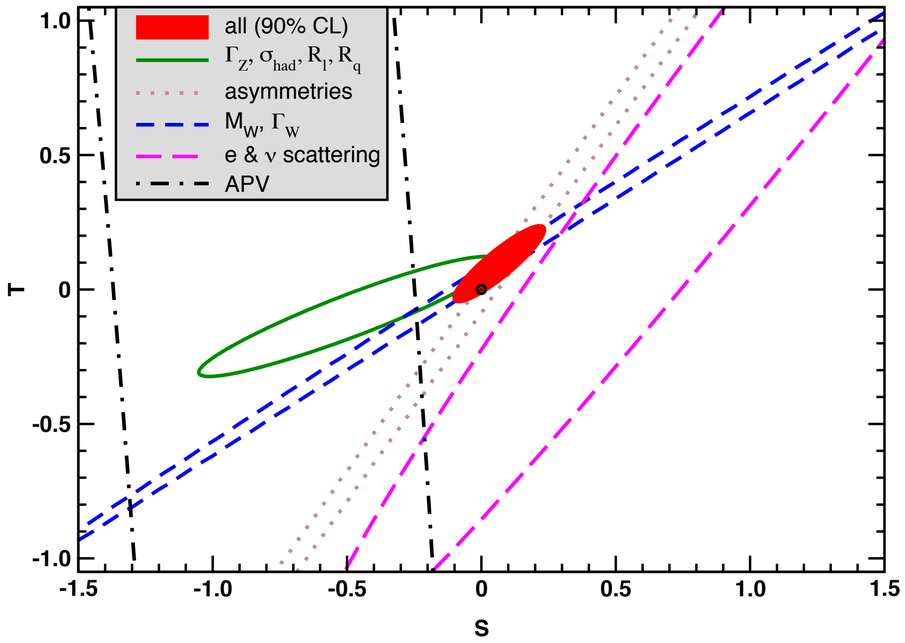
New radiative corrections



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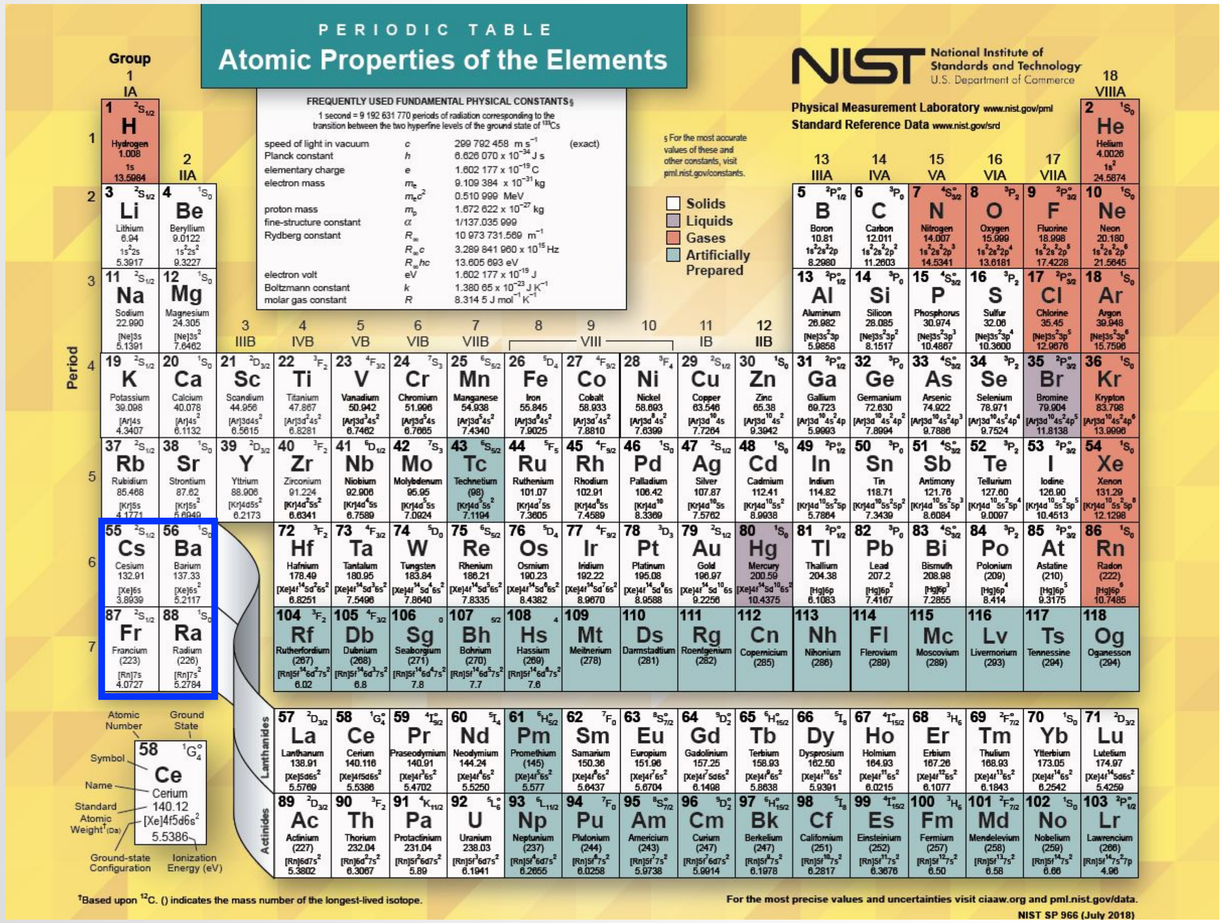
$$Q_W - Q_W^{\rm SM} = -0.800\, S - 0.007\, T$$

T vs S



Erler and Freitas, Particle Data Group review (2016)
(new APV results now centred around SM value)

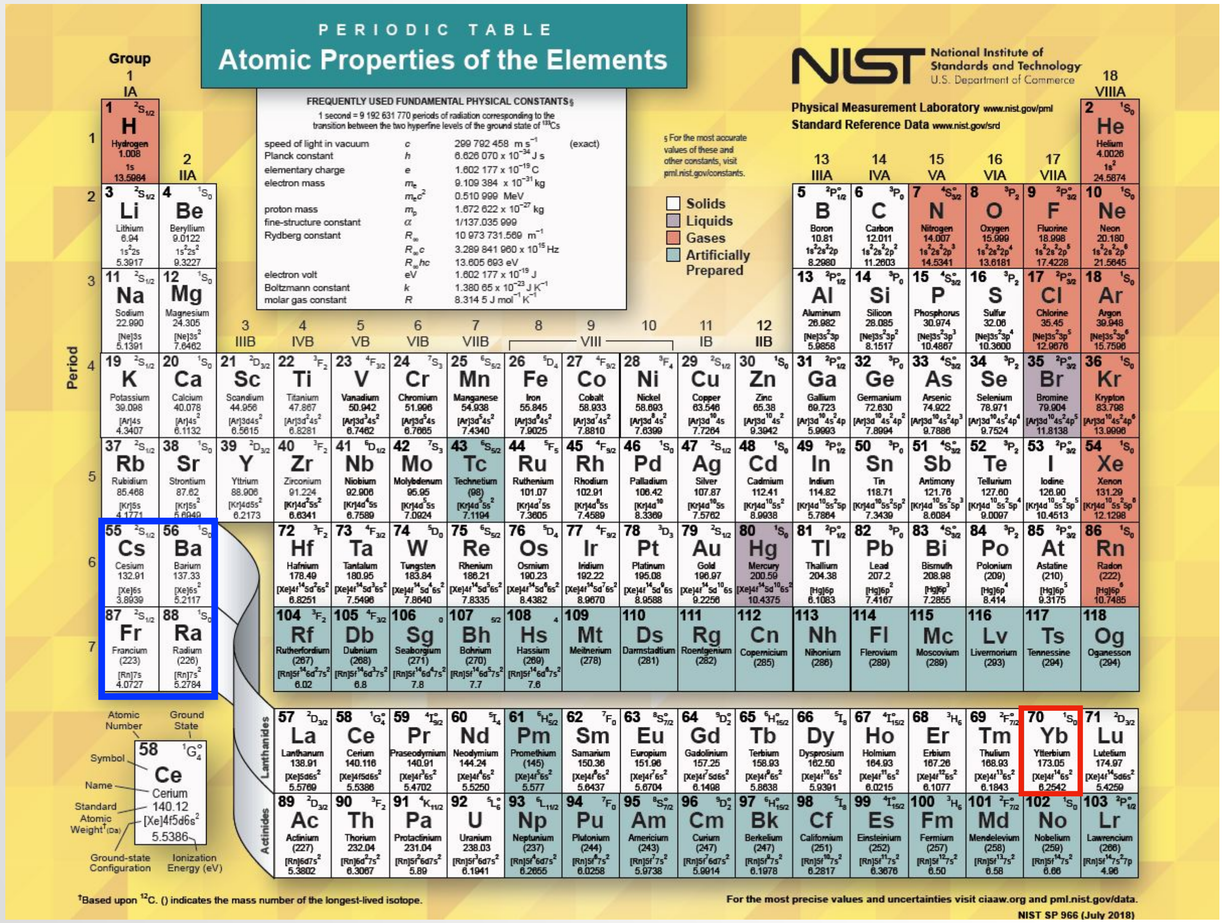
Experiments in preparation/progress



Neutral atoms: Cs (Purdue) ; Fr (TRIUMF; Tokyo)

Singly-ionized atoms: Ba^+ (Seattle) ; Ra^+ (Groningen)

Experiments in preparation/progress



Neutral atoms: Cs (Purdue) ; Fr (TRIUMF; Tokyo)

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- Our *precision atomic theory* group at UQ is aiming towards 0.1% uncertainty in APV calculations

Related studies:

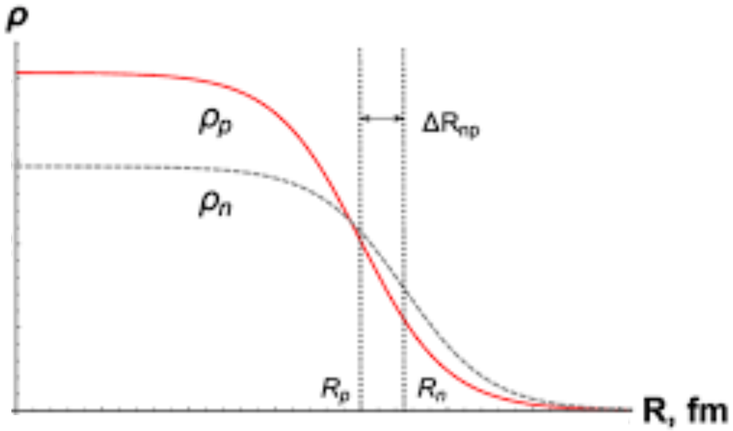
- Atomic parity violation along an isotope chain

— *remove the dependence on atomic theory*

— brings in dependence on *neutron skin*

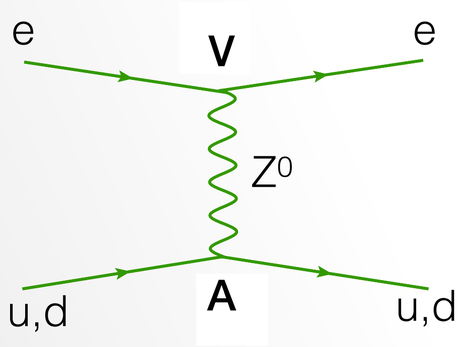
Measurements in Yb (Mainz)

Antypas *et al.*, Nat. Phys. (2019)



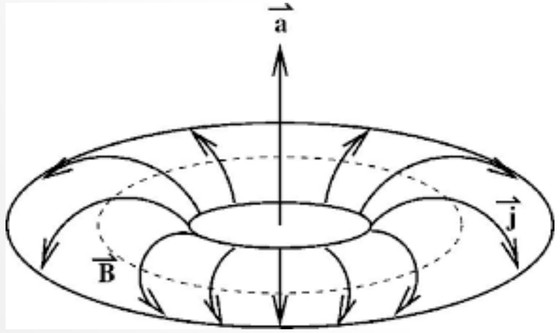
Anapole

What about:



- Leads to *nuclear-spin-dependent* effects in atoms

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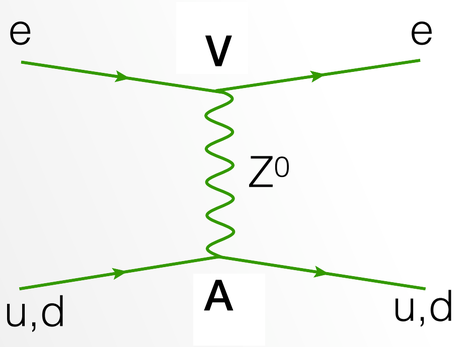


- Measured with 14% uncertainty in Cs!

Haxton and Wieman, Ann. Rev. Nuc. Part. Sci. (2001)

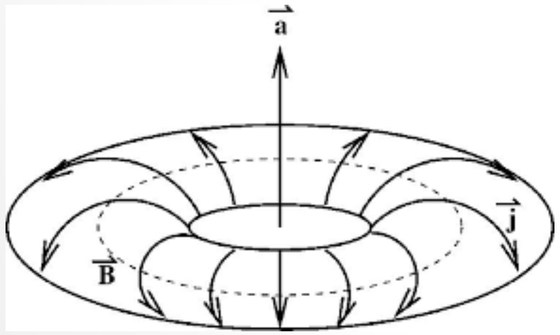
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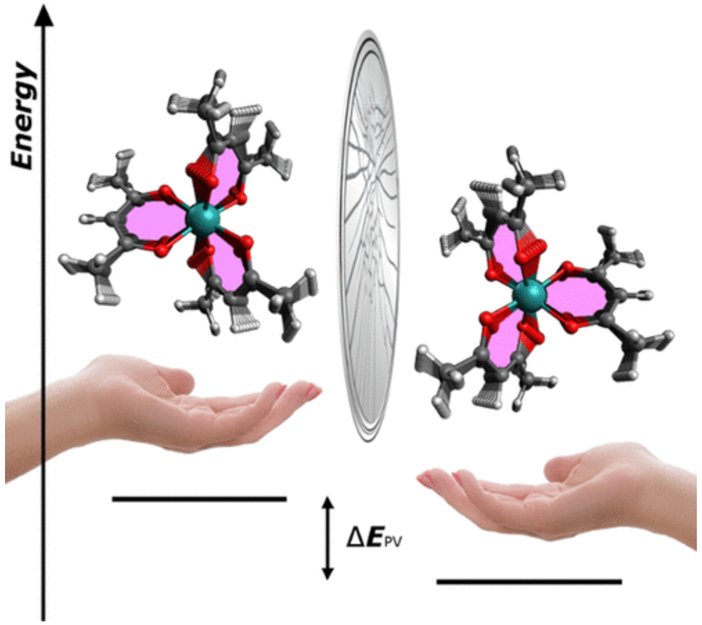
To consider

Homochirality of biological molecules:

DNA is right-handed;

Amino acids are left-handed;

Sugars are right-handed.



Could the weak interaction play a role in the preferred production of molecules of one handedness over another?

Resources:

Ginges and Flambaum, *Violations of fundamental symmetries and tests of unification theories of elementary particles*, Phys. Rep. 397, 63 (2004)

Safronova *et al*., *Search for new physics with atoms and molecules*, Rev. Mod. Phys. **90**, 025008 (2018)

Ramsey-Musolf, *Low-energy parity violation and new physics*, Phys. Rev. C. **60,** 015501 (1999)

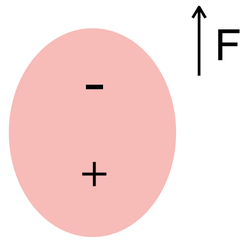
Khriplovich, Parity nonconservation in atomic phenomena, Phys. Rev. C. **60,** 015501 (Gordon and Breach, Philadelphia, 1991)

Time-reversal violating electric dipole moments

Electric dipole moments (EDMs)

Intrinsic EDMs of quantum systems (e.g., elementary particle, neutron, atom) violate parity P and time-reversal T

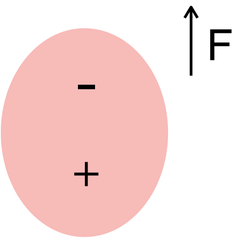
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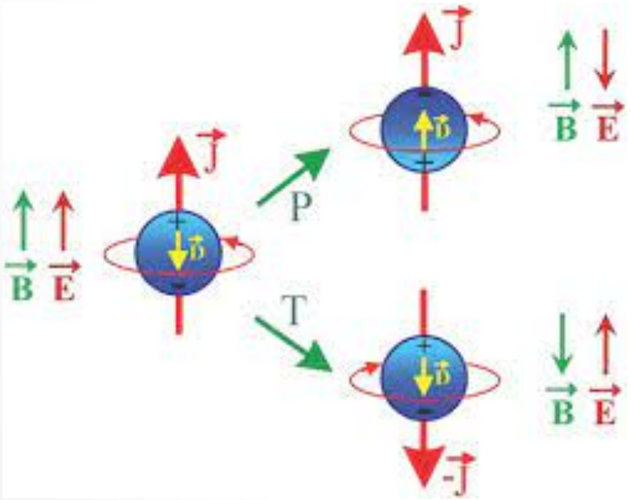


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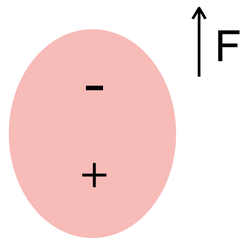




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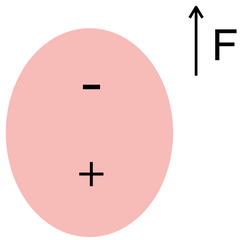
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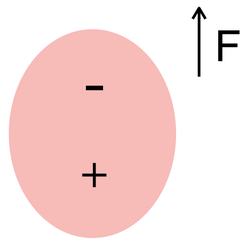
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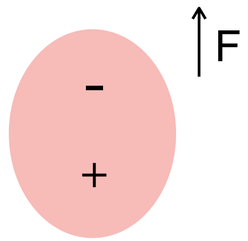
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- Not enough CP-violation in SM to explain matter-antimatter asymmetry of universe!



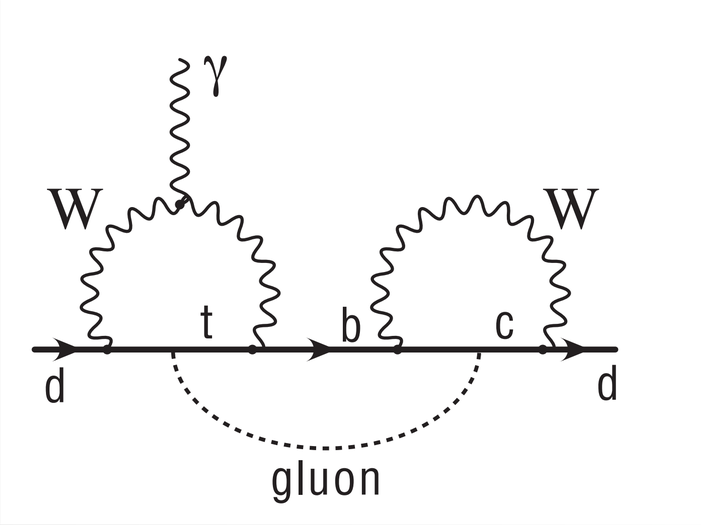
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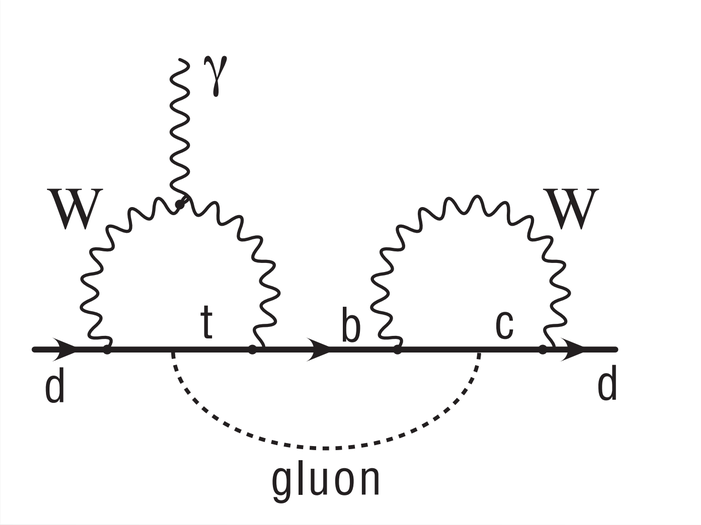


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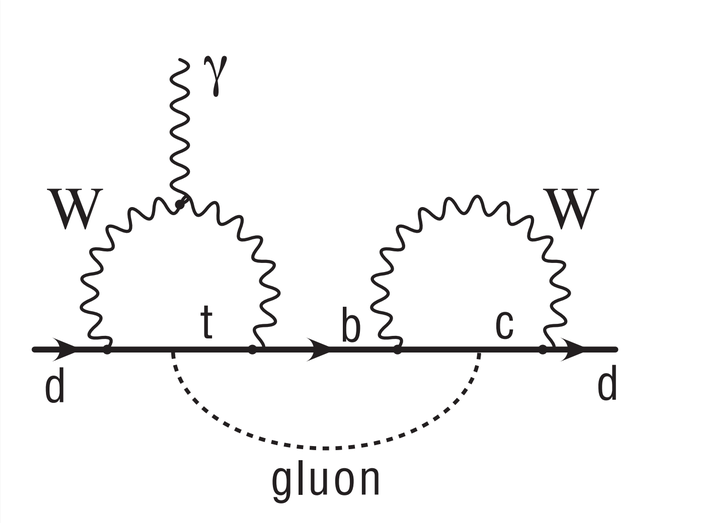


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Phenomenology

Phenomenological description, effective operators

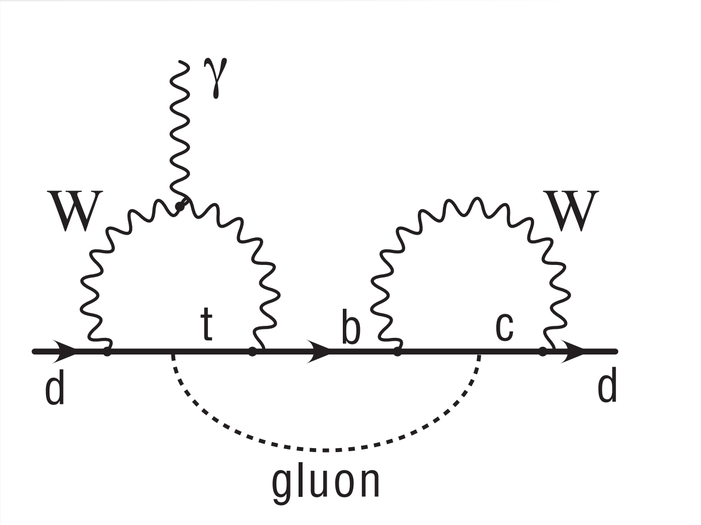
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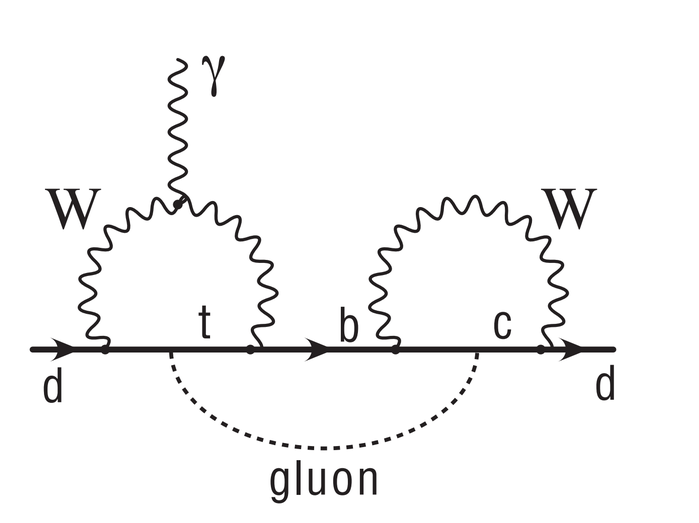
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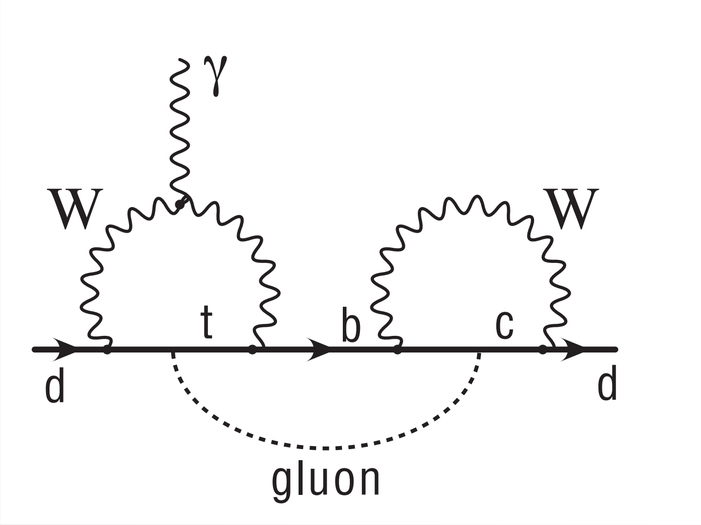
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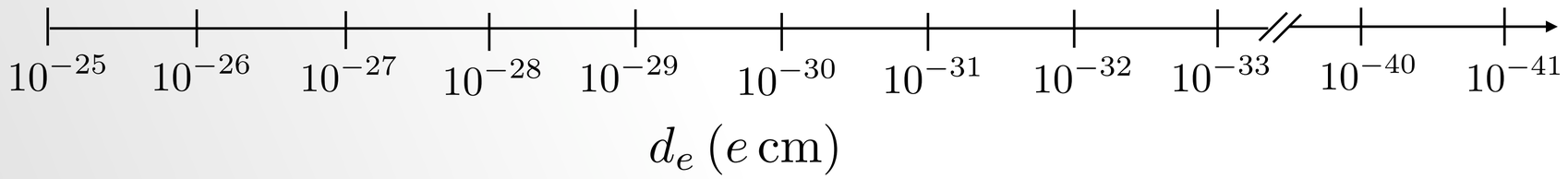
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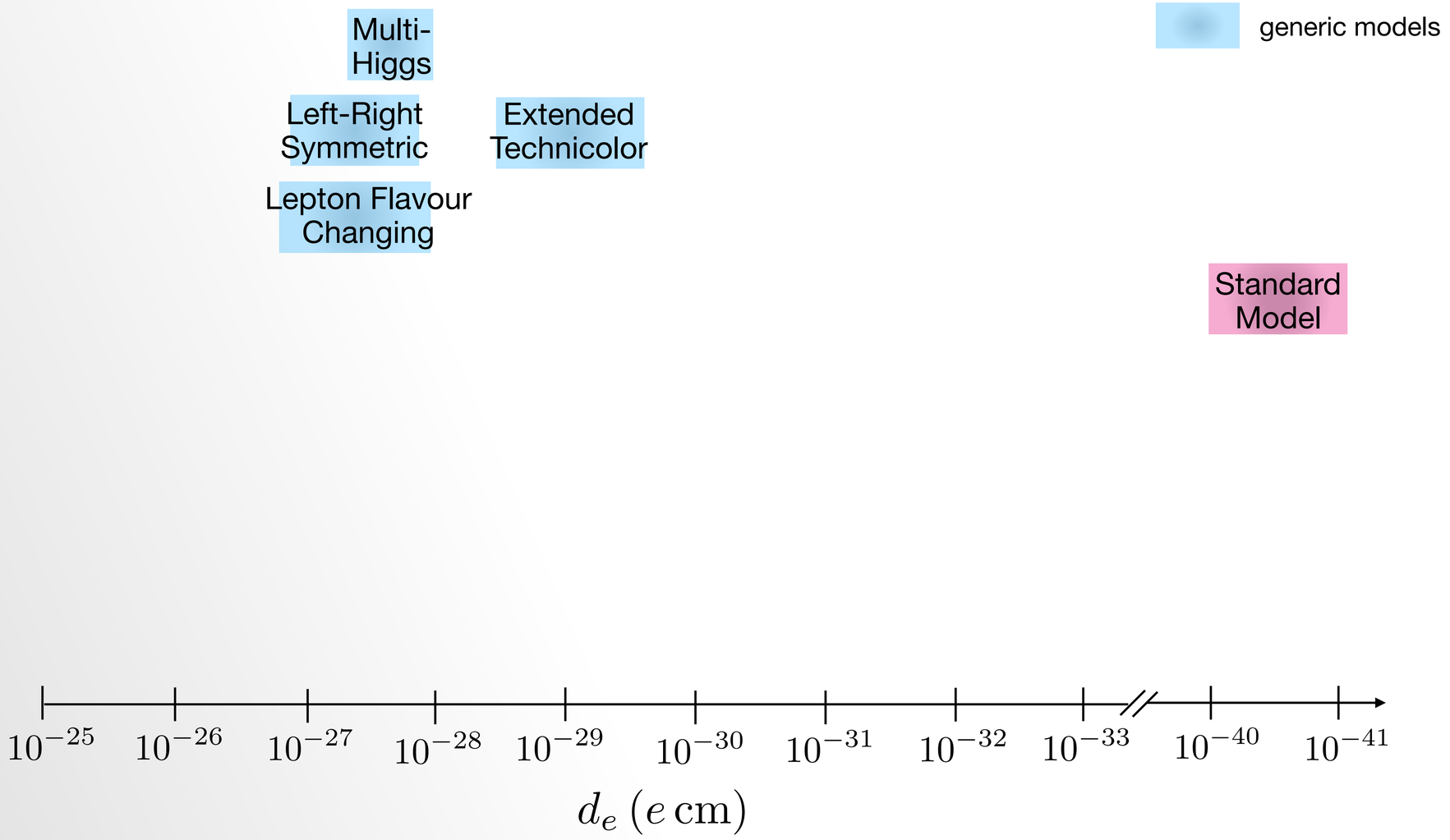
Predictions and bounds — electron EDM

Standard Model



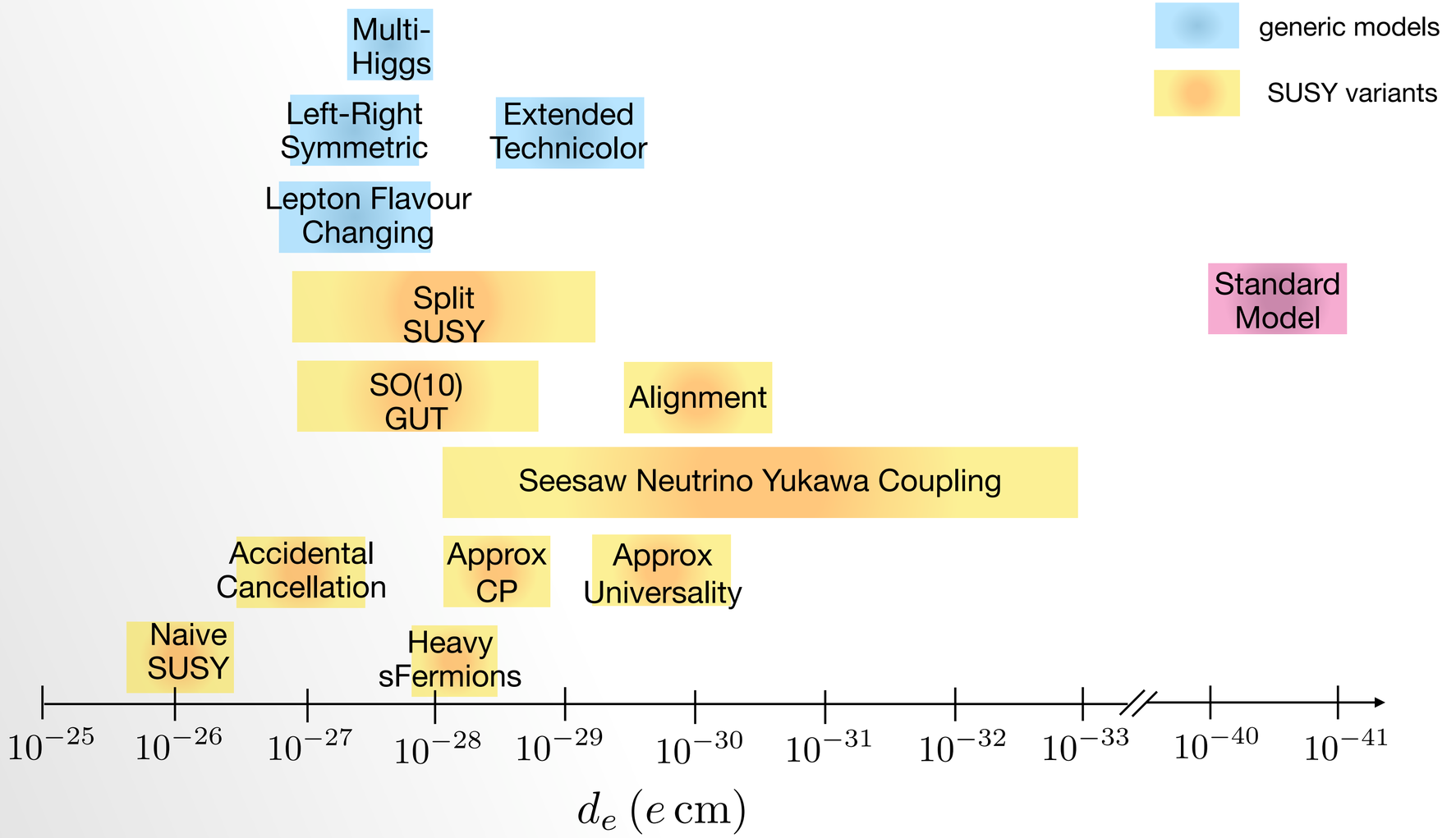
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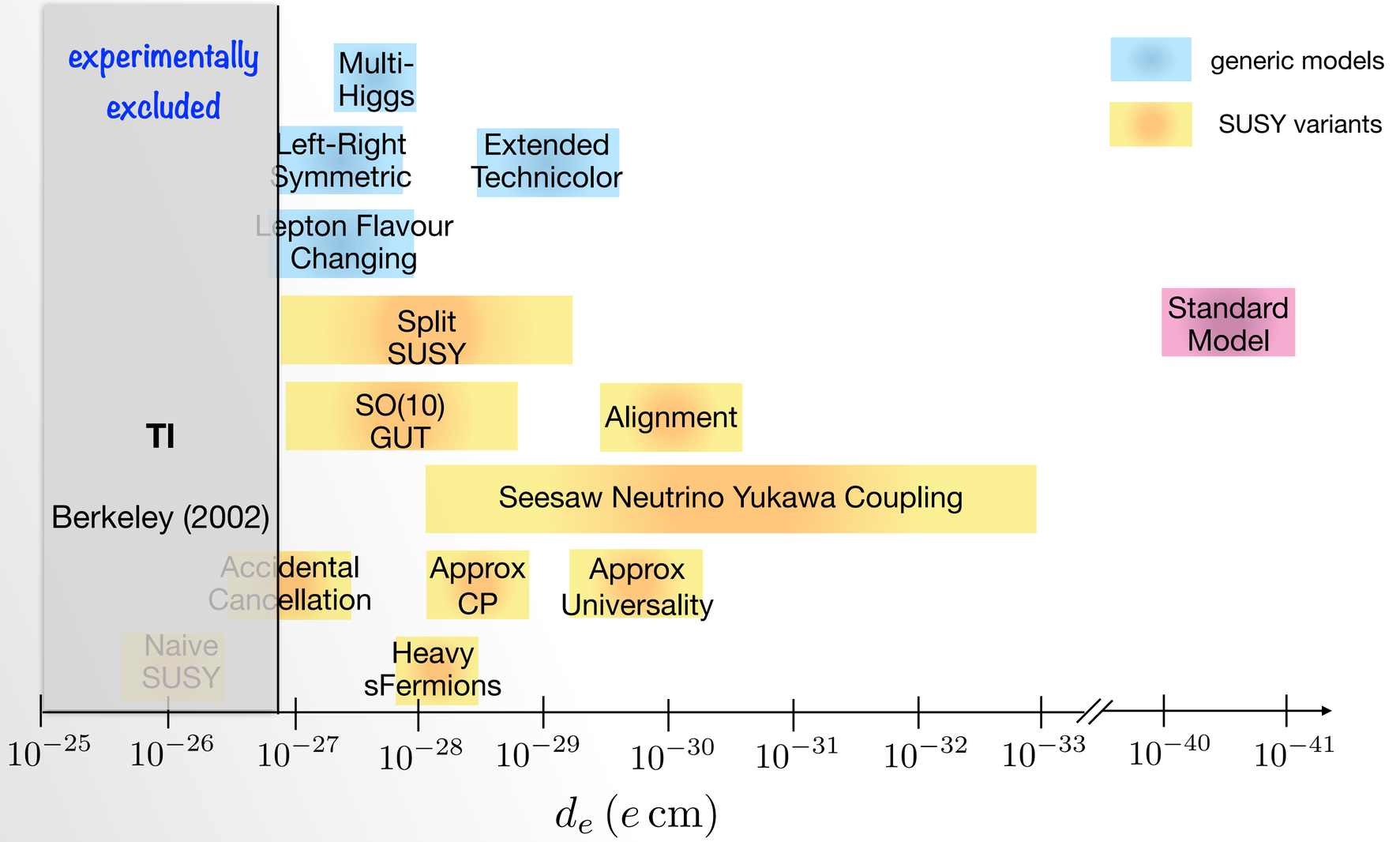
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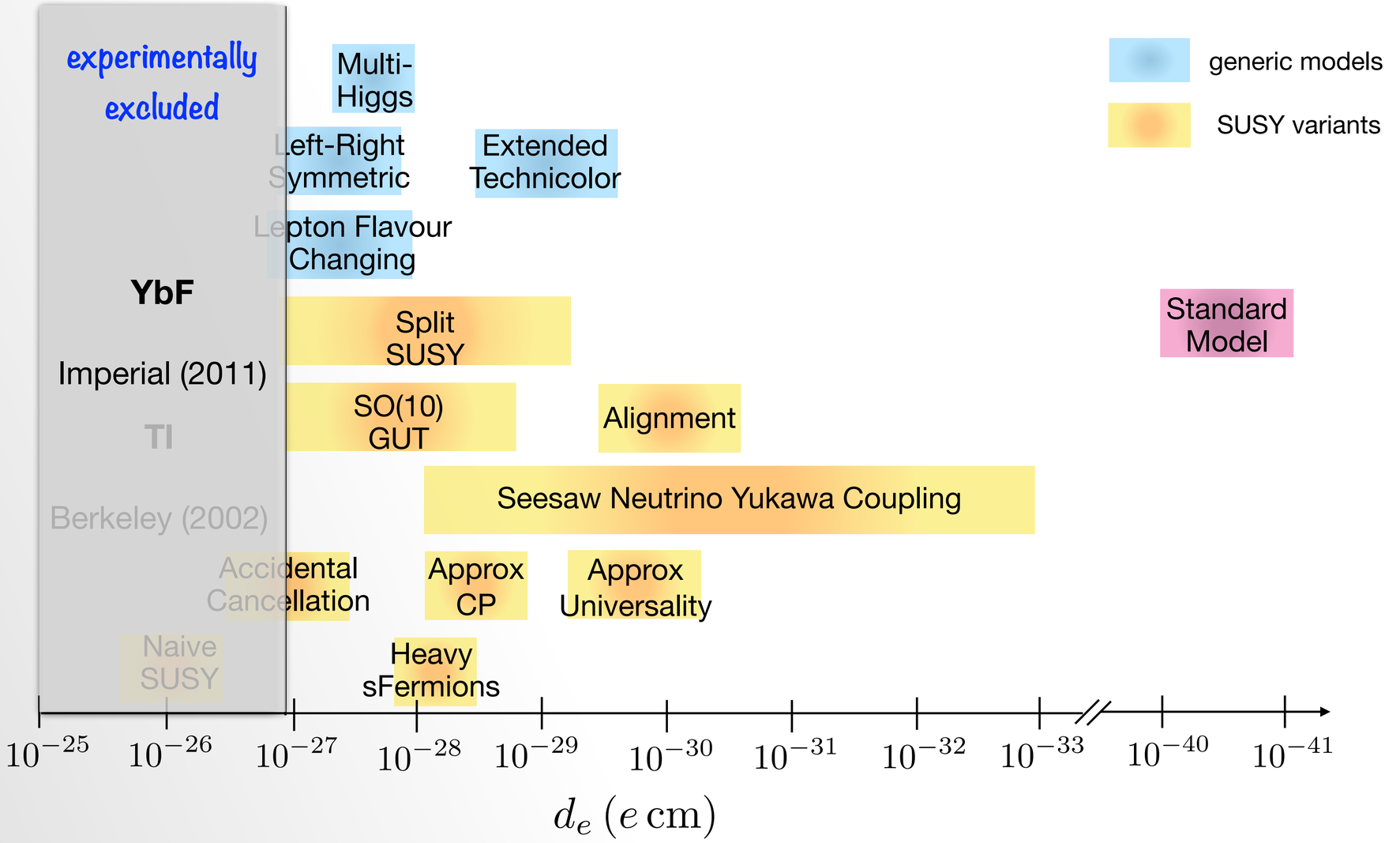
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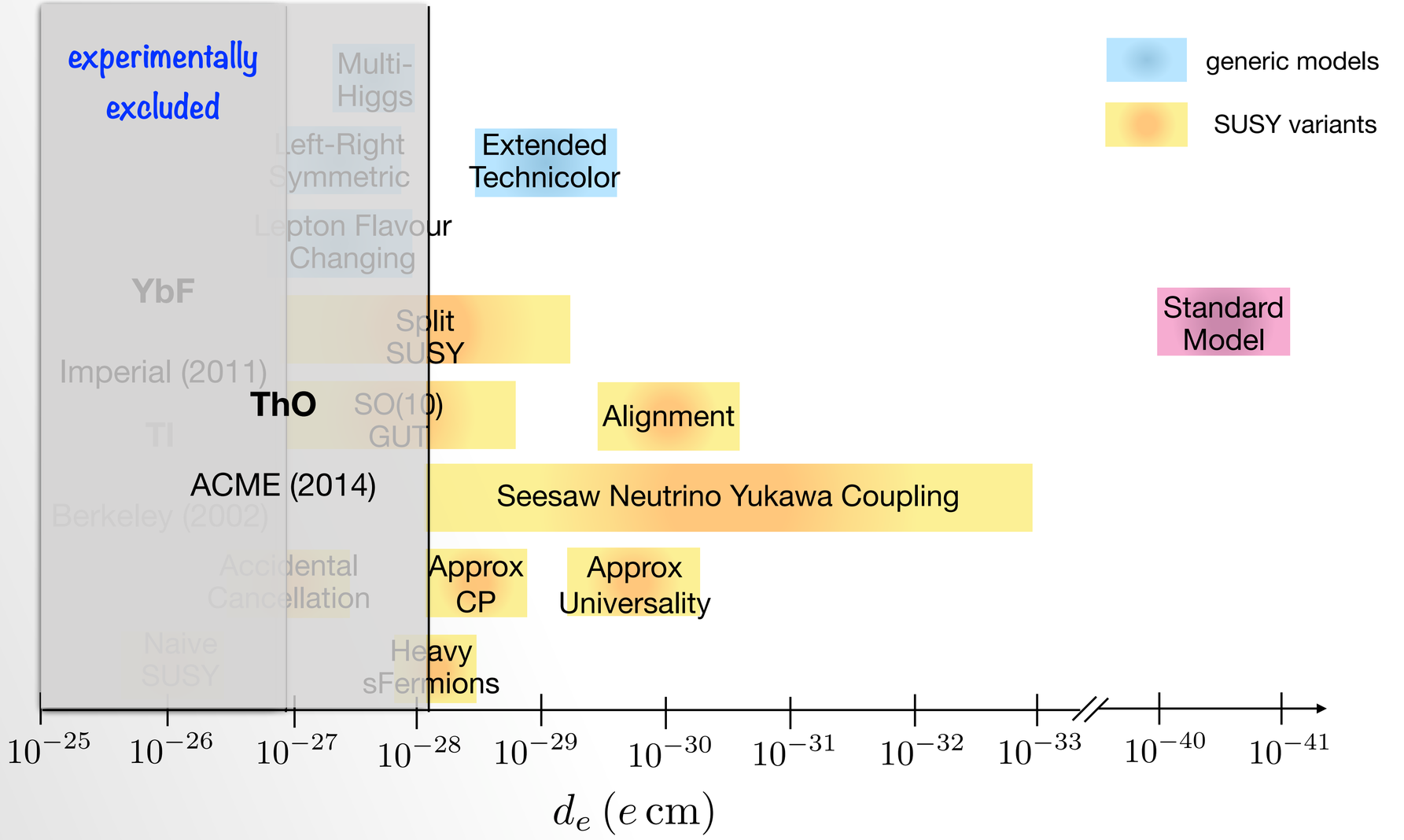
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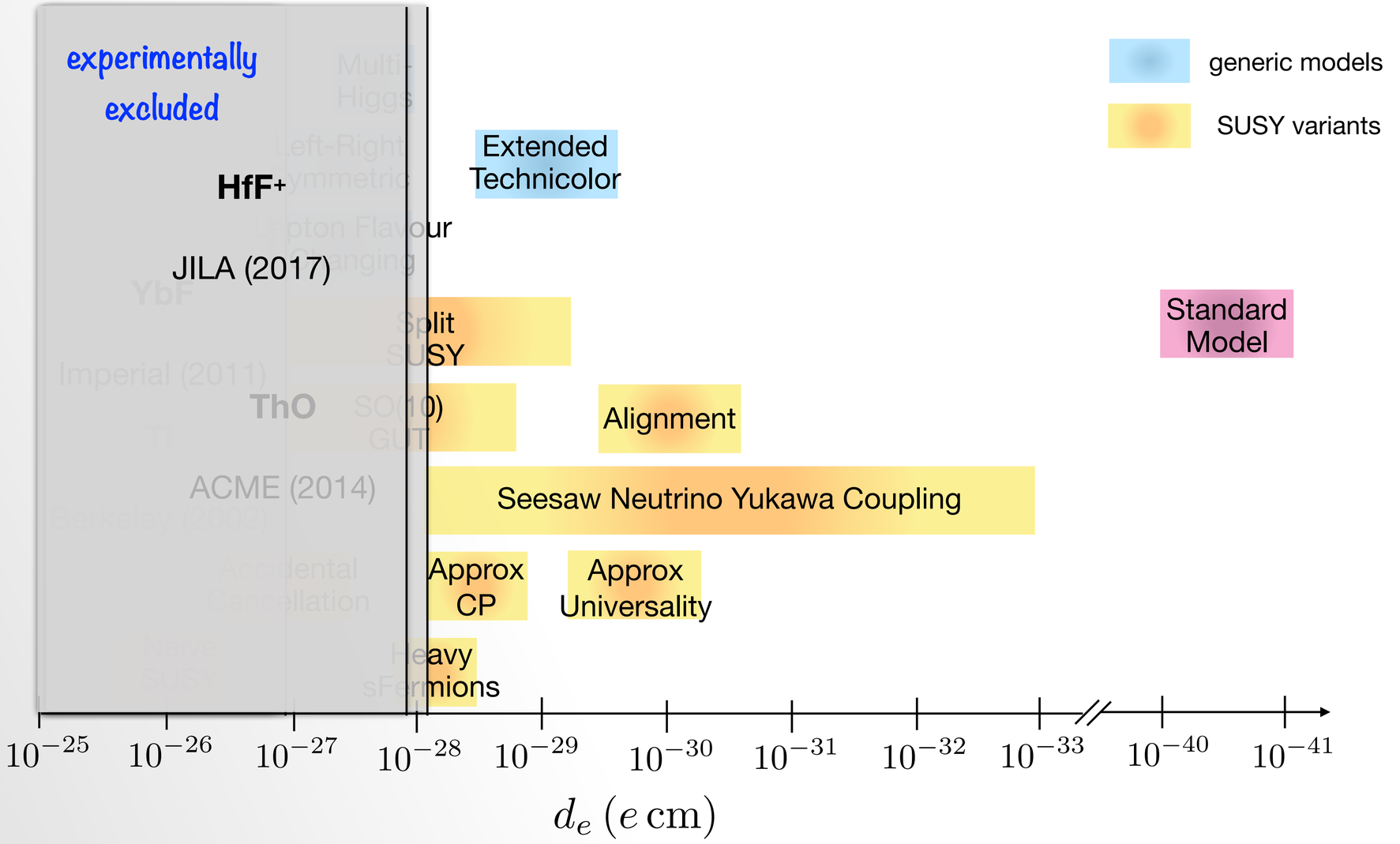
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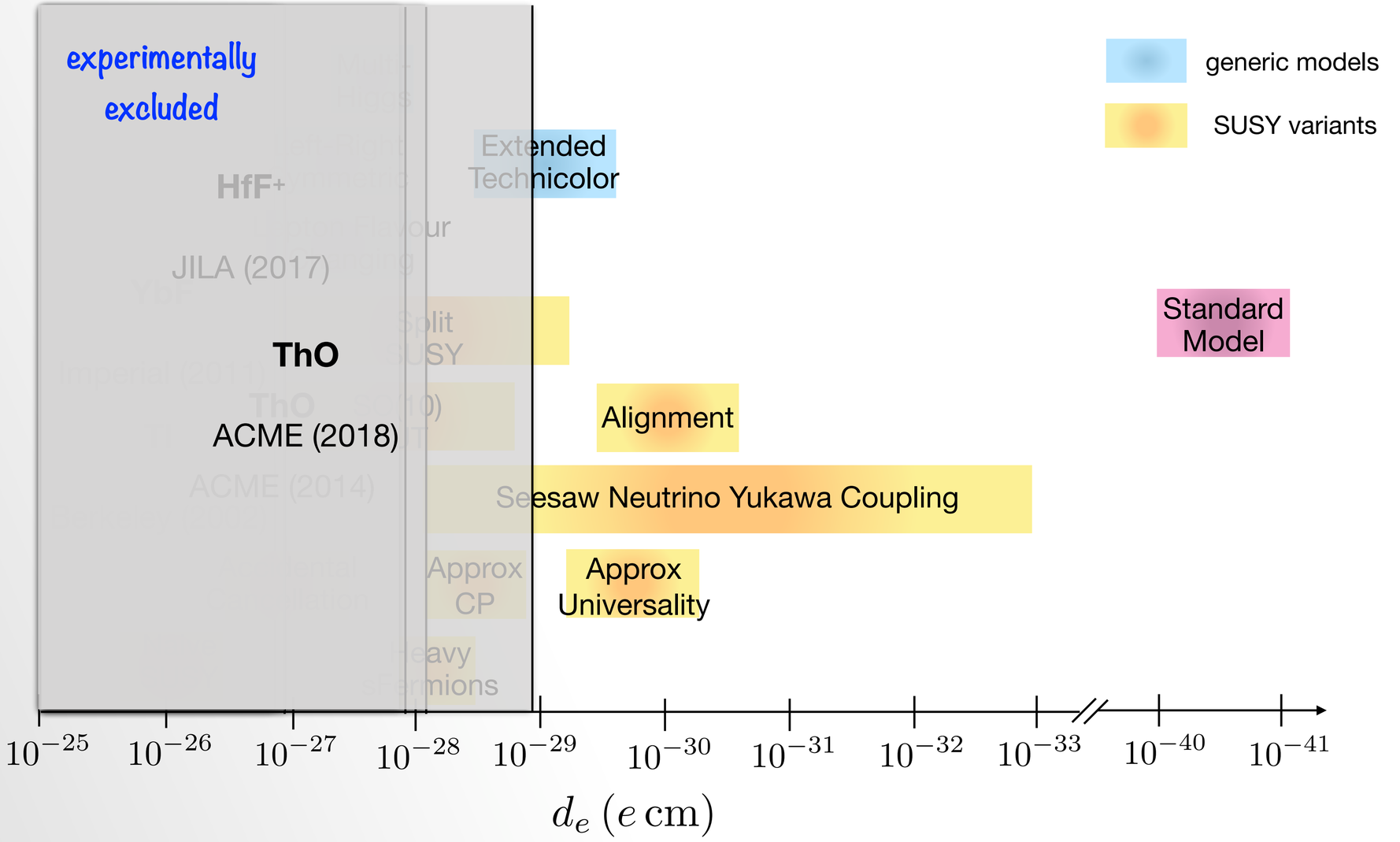
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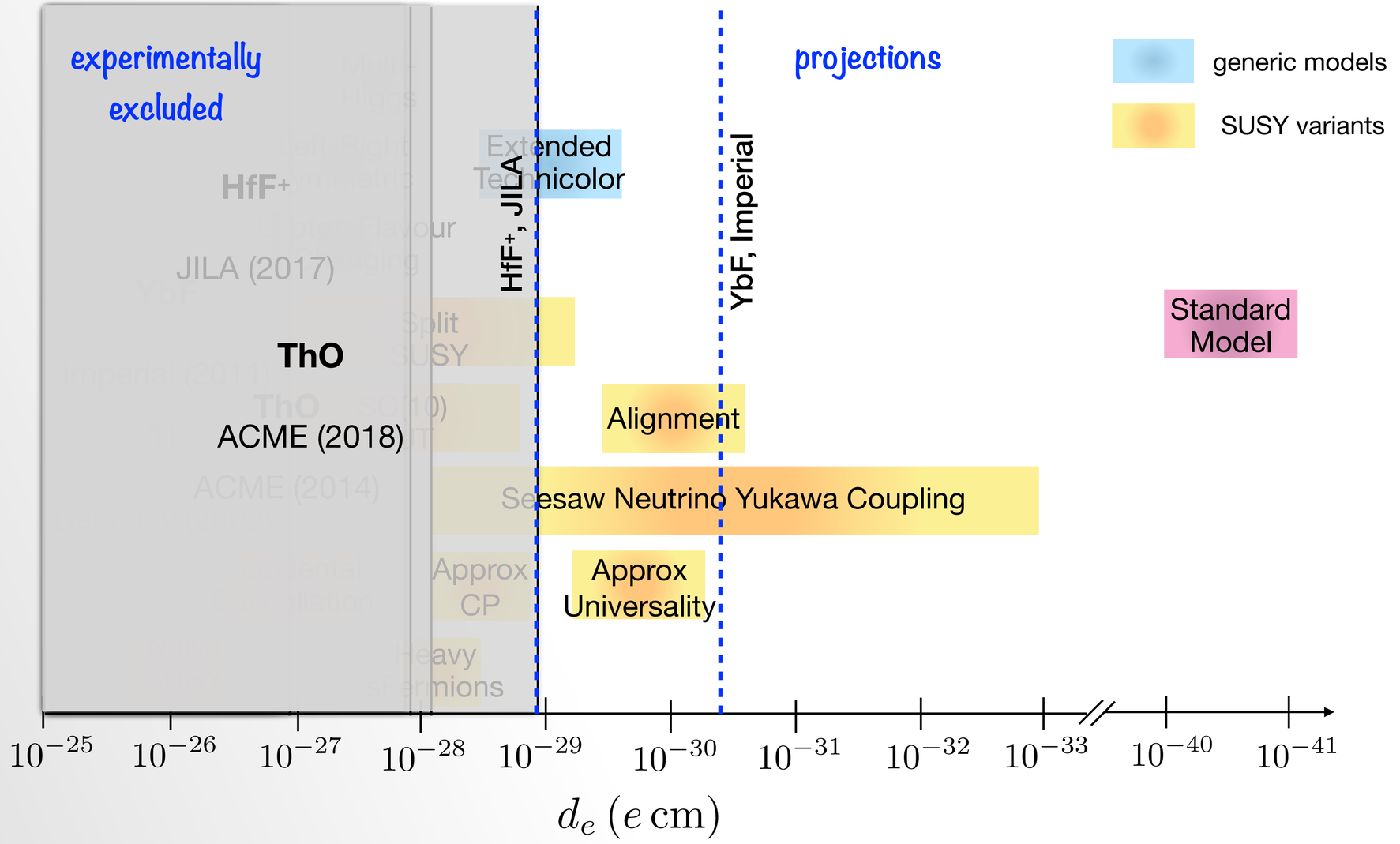
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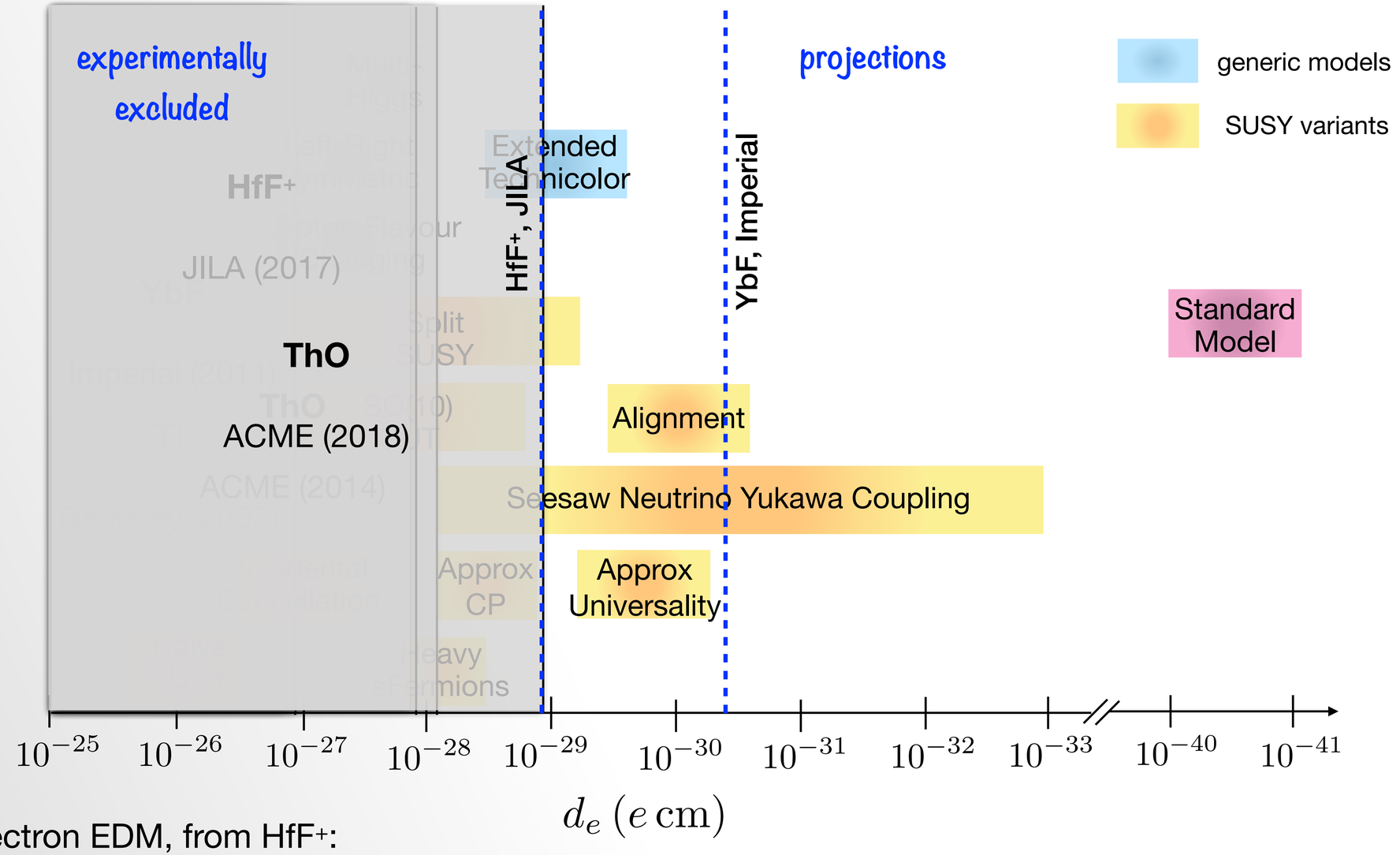
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ACME collaboration — Harvard, Yale, Northwestern, Caltech

Predictions and bounds — electron EDM



Best limit on electron EDM, from HfF+:

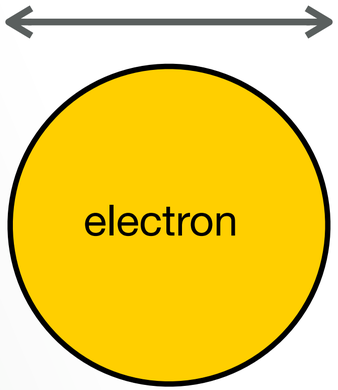
$$|d_e| < 4.1 \times 10^{-30}\, e\,\mathrm{cm}$$ [Roussy et al., Science (2023)]

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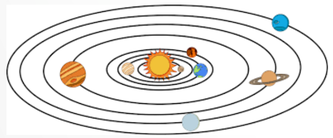
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Scale of precision

Q. If the electron were the size of the solar system, it has been measured to be perfectly “round” with uncertainty:



solar system

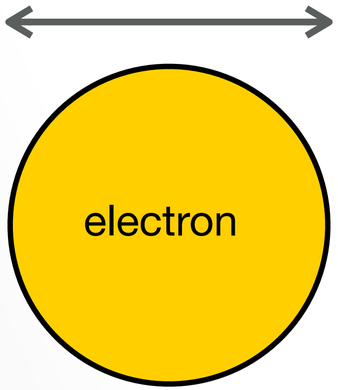


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B. The size of Earth.
C. The size of a football field.
D. A fraction of the width of a human hair.

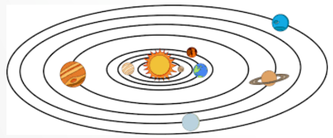
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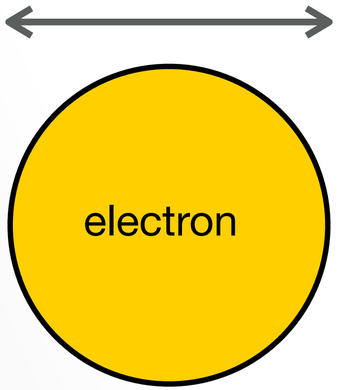


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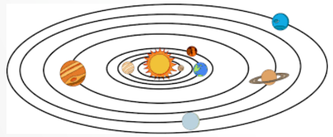
[Ed Hinds]

Scale of precision

Q. If the electron were the size of the solar system, it has been measured to be perfectly "round" with uncertainty:



solar system



A. The size of the sun.
B. The size of Earth.
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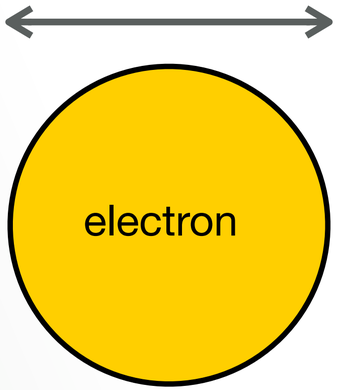
Energy reach

- EDM of fermion of mass *m* arising due to new heavy particle of mass Λ and phase $\phi_{\rm CP}$ in diagram with *n* loops:

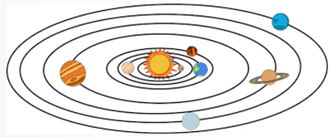
$$d \sim \mu \, \sin\phi_{\rm CP} \, (g^2/2\pi)^n \, (m/\Lambda)^2$$

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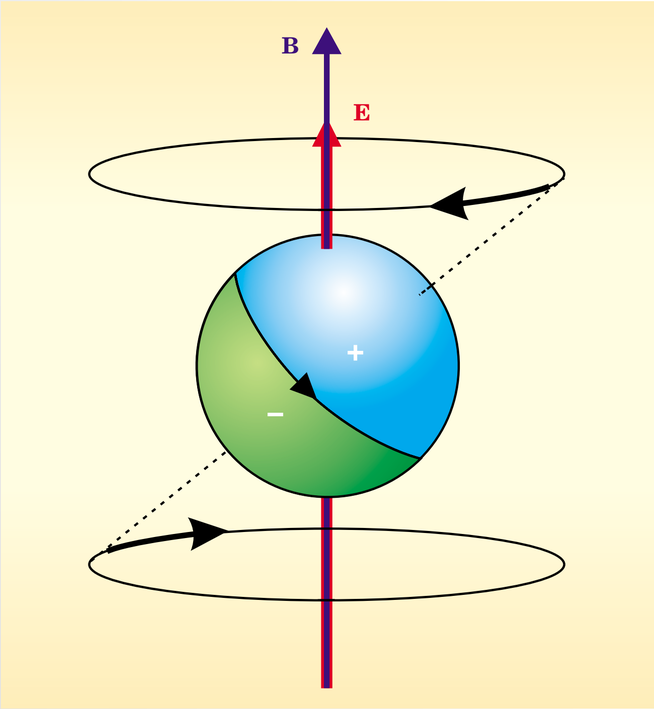
- Current bounds from atomic and molecular experiments give

$$\Lambda \gtrsim (1 - 1000\,\text{TeV})\sqrt{\sin\phi_{\rm CP}}$$

[Ramsey-Musolf]

Basics of an EDM measurement

- Measure difference in precession frequencies in parallel and antiparallel electric and magnetic fields

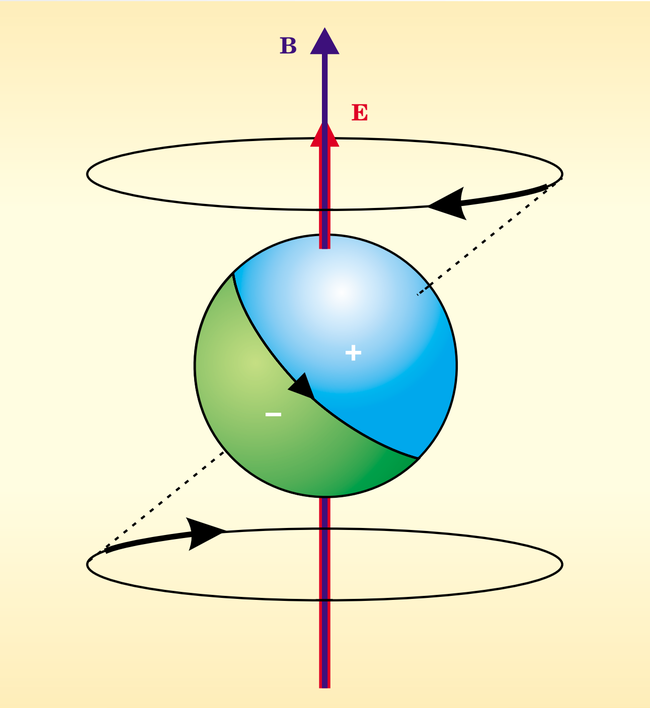


$$H = -\mu \cdot \mathbf{B} - \mathbf{d} \cdot \mathbf{E}$$

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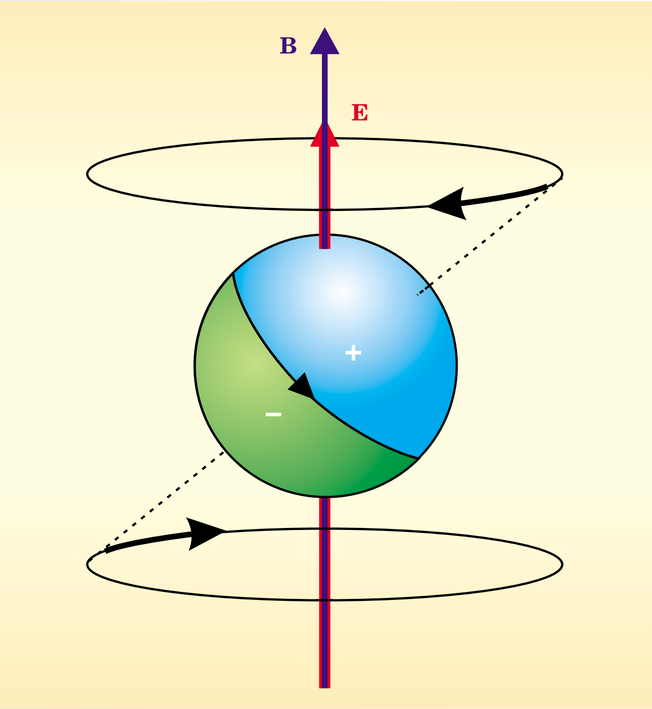
Atomic EDMs

- P,T-violating interactions induce EDMs in atoms

$$\mathbf{d}_{\mathrm{atom}} = \langle \tilde{N}|\mathbf{D}|\tilde{N}\rangle = 2\sum_{M} \frac{\langle N|\mathbf{D}|M\rangle\langle M|H_{PT}|N\rangle}{E_N - E_M} = d_{\mathrm{atom}}\,\mathbf{F}/F$$

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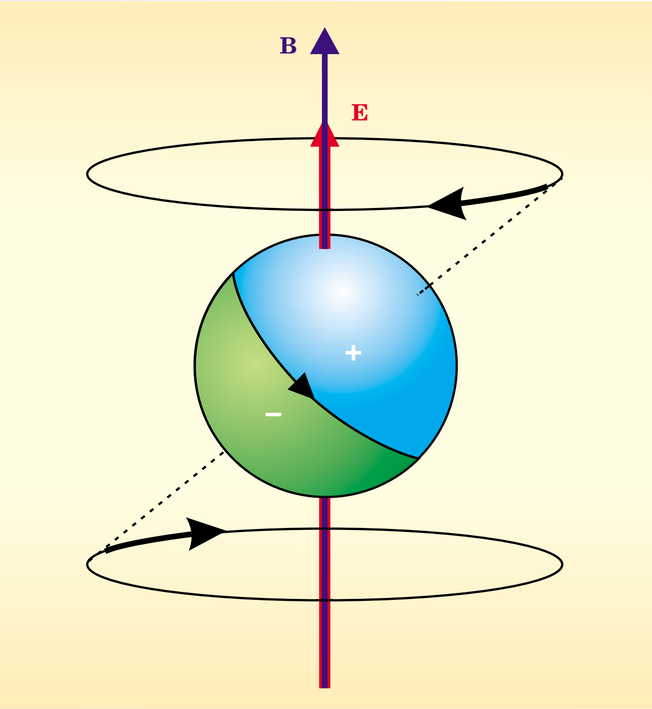
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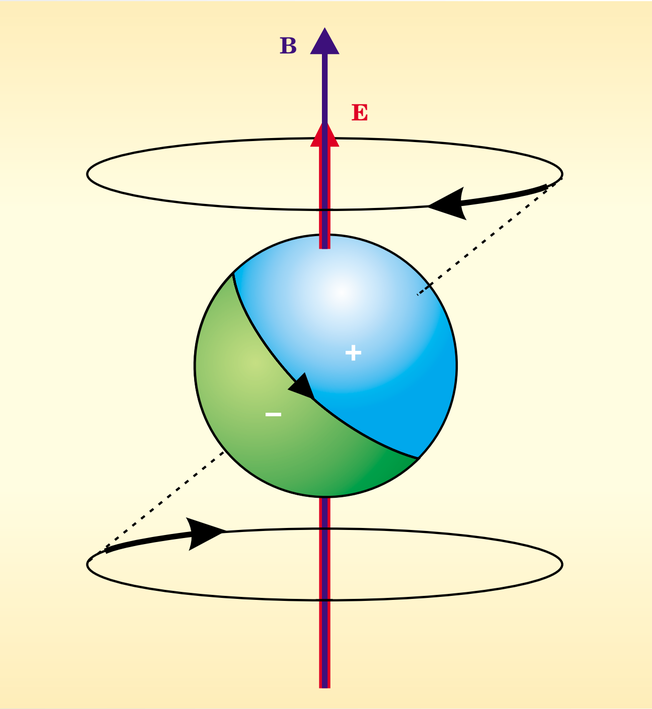
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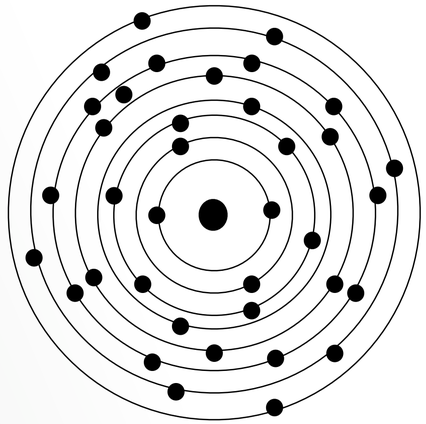
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Diamagnetic systems:
hadronic and semi-leptonic mechanisms

Paramagnetic systems:
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Schiff screening

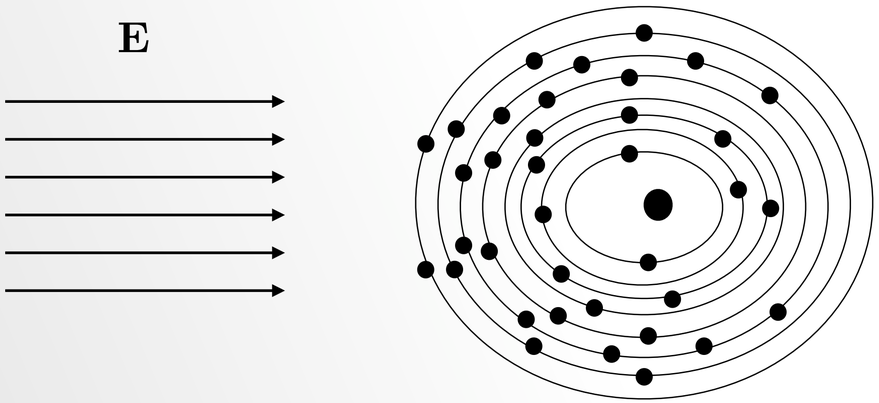
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Consider atom made of point charges with electric dipole moments and experiencing Coulomb interaction only. Treat non-relativistically.

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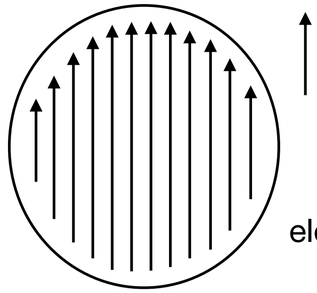
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Flambaum, Ginges (2002)

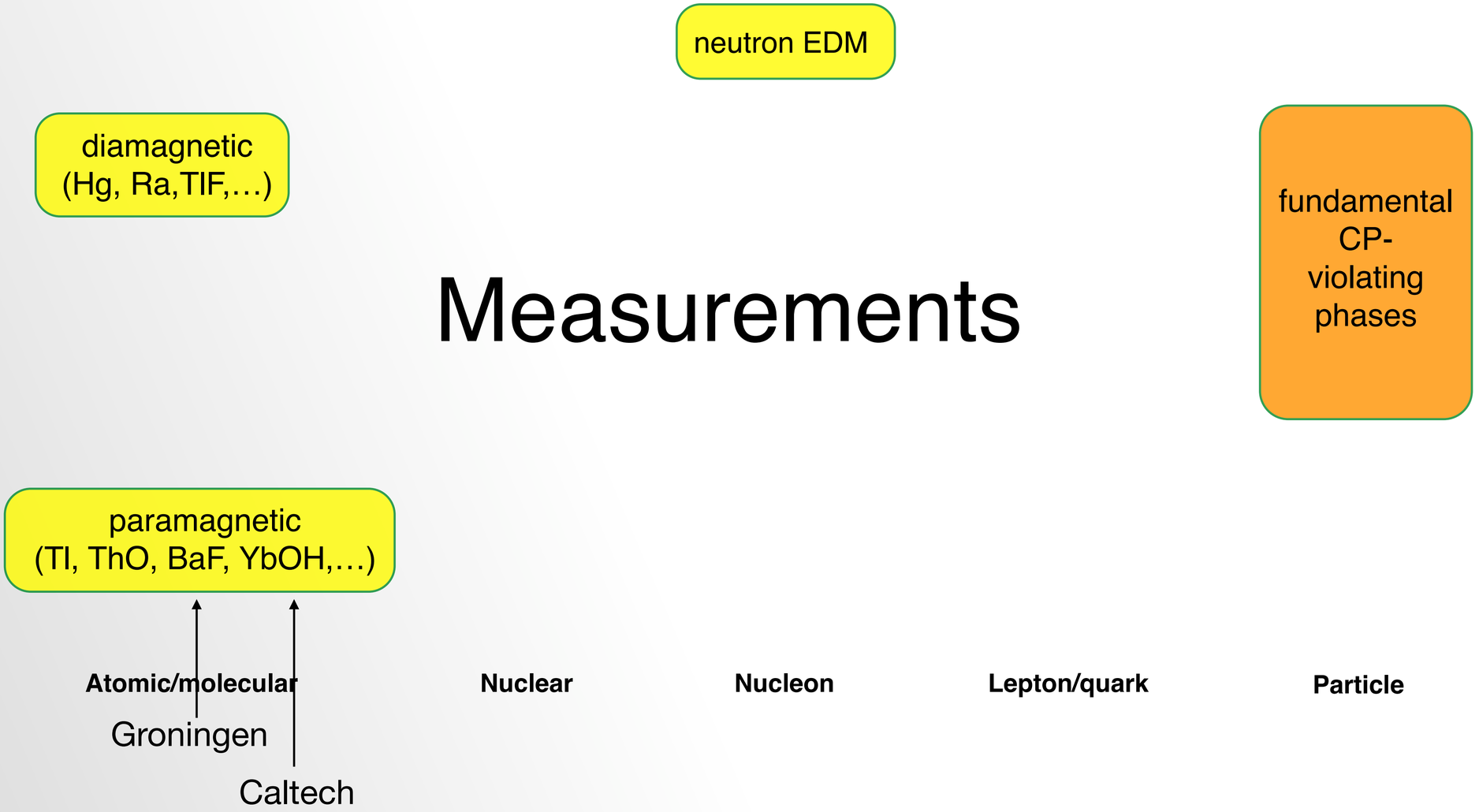
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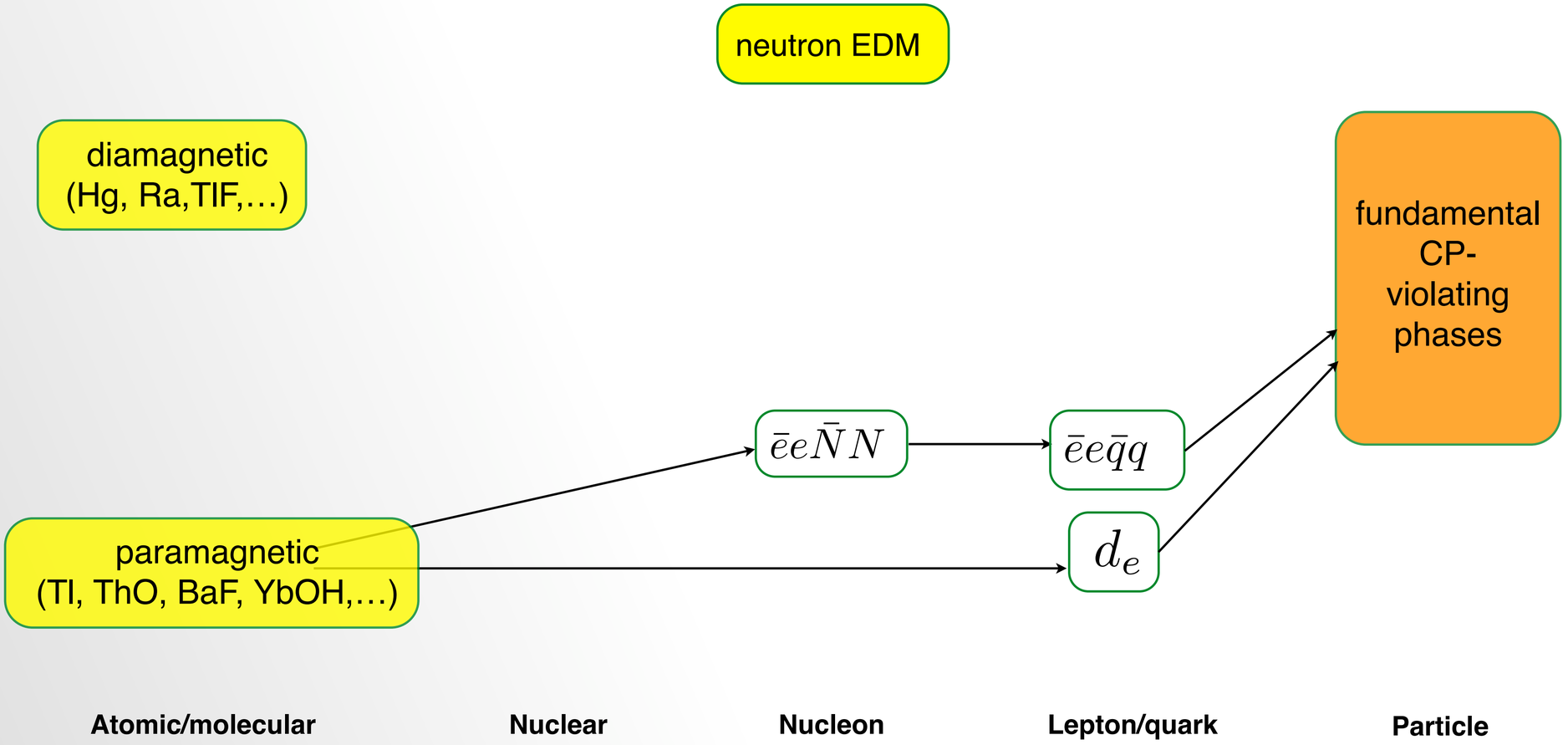
Finite nuclear size ⇒ nuclear Schiff moment! $d_{\rm atom} = \eta \, S$, $\mathbf{S} = S\mathbf{I}/I = \frac{e}{10} \left[\langle r^2 \mathbf{r} \rangle - \frac{5}{3Z} \langle r^2 \rangle \langle \mathbf{r} \rangle \right]$

Relativistic effects ⇒ electron EDM, with enhancement! $d_{\rm atom} = K \, d_e$, $|K({\rm Tl})| \approx 600$

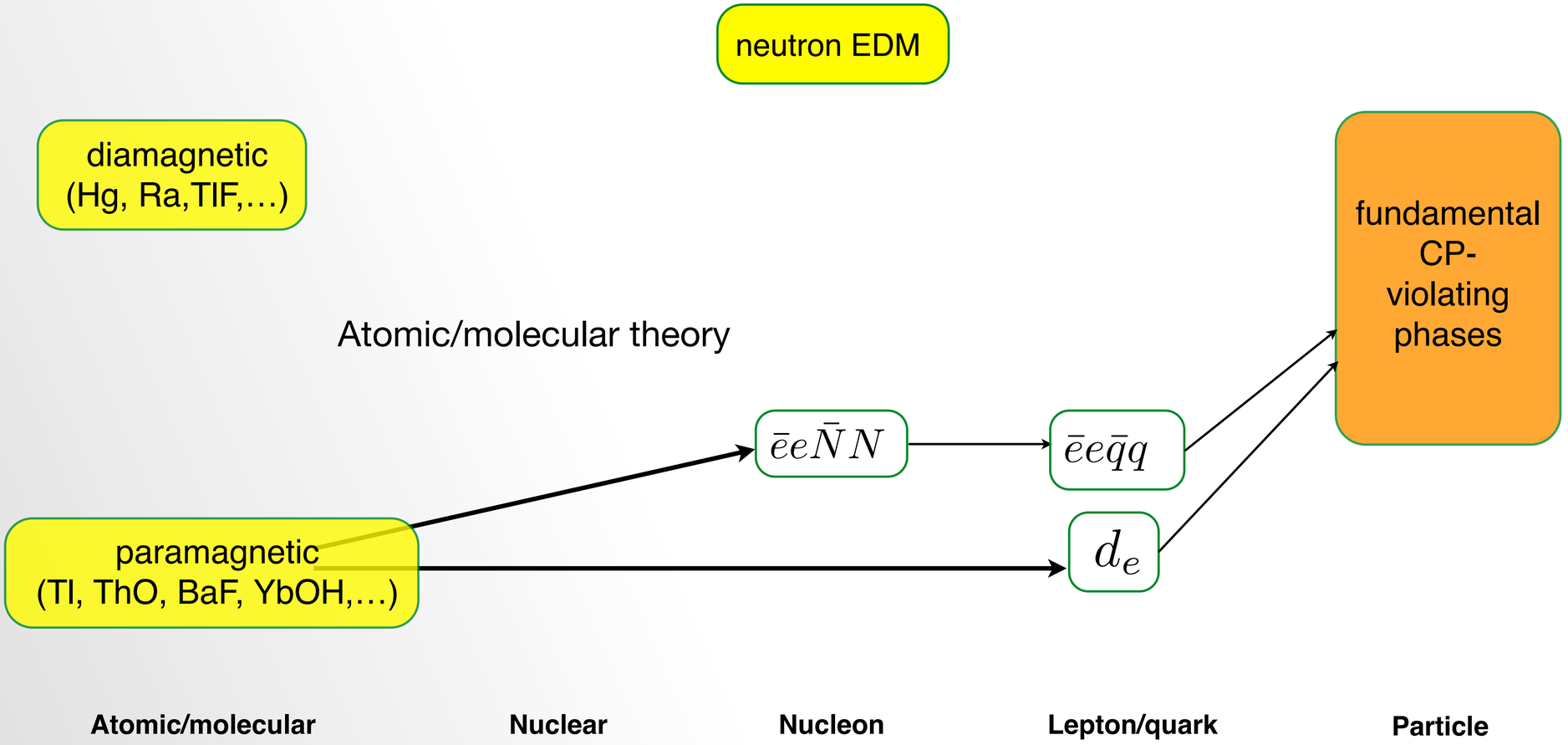
Leading mechanisms



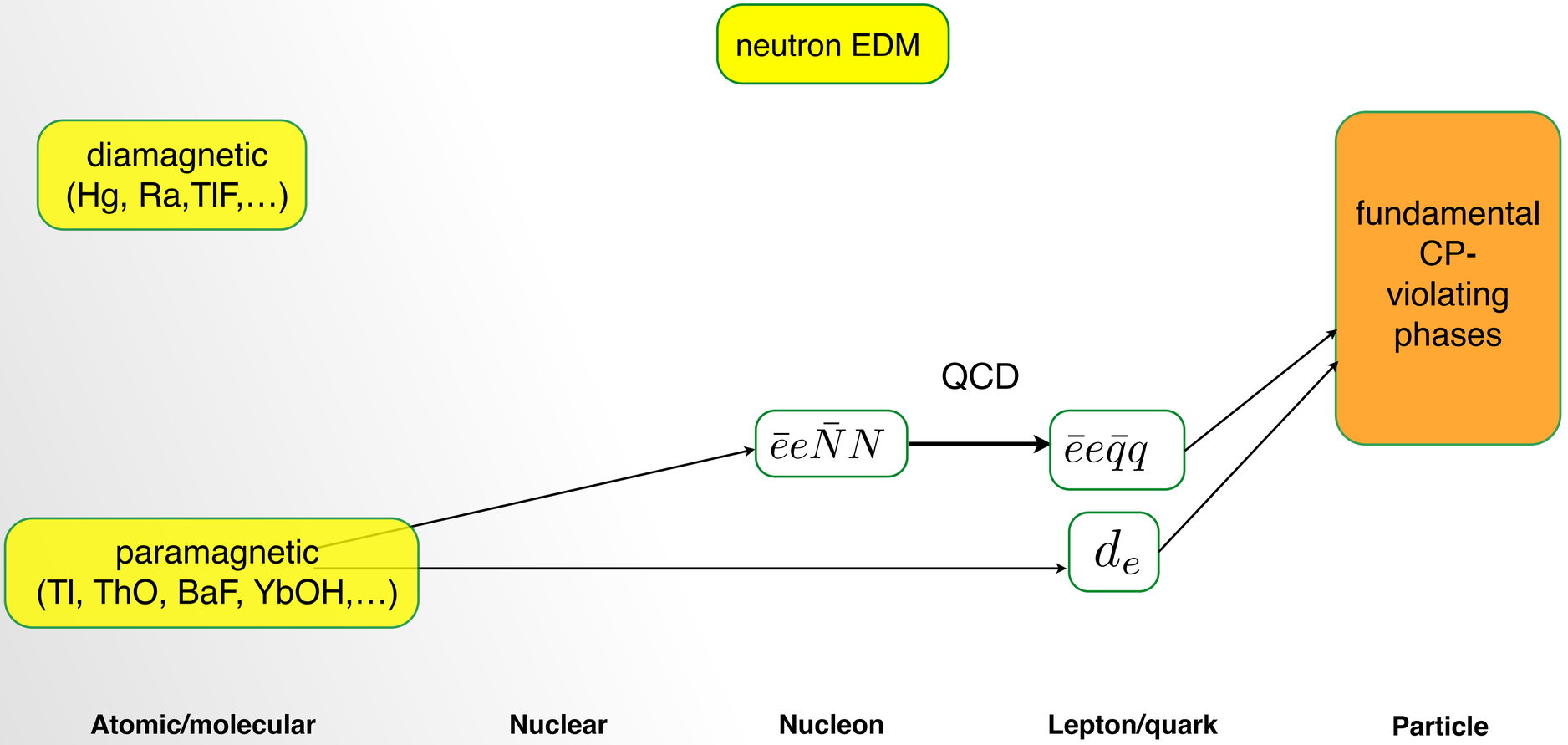
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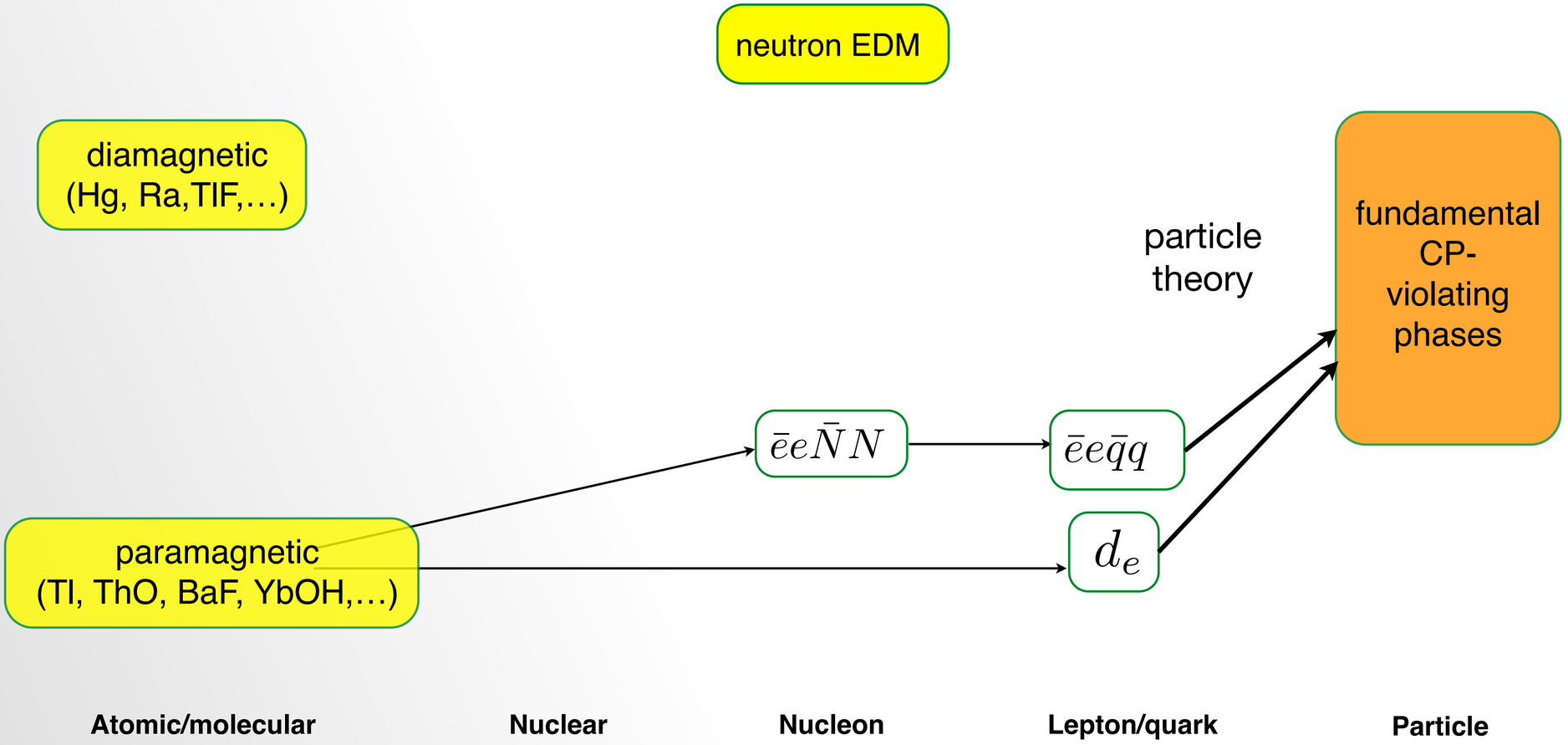
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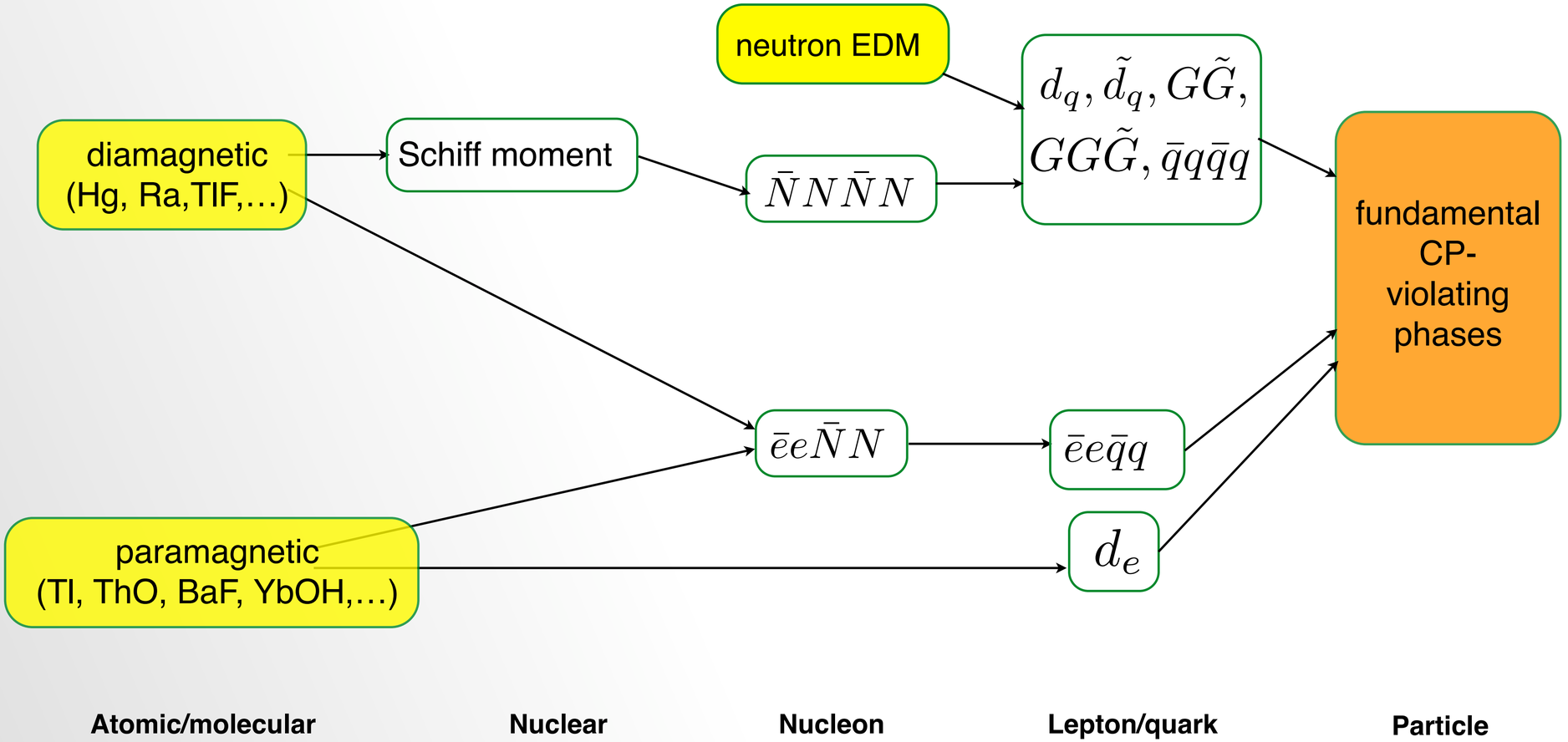
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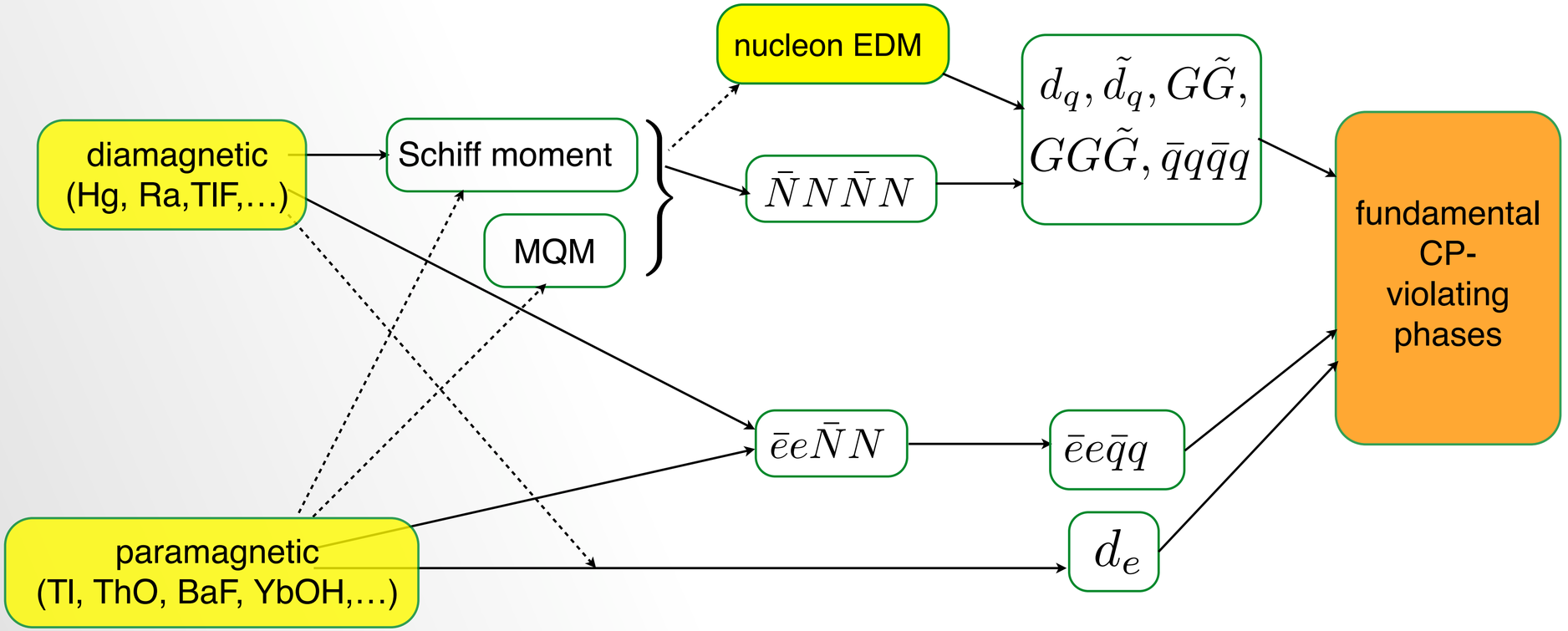
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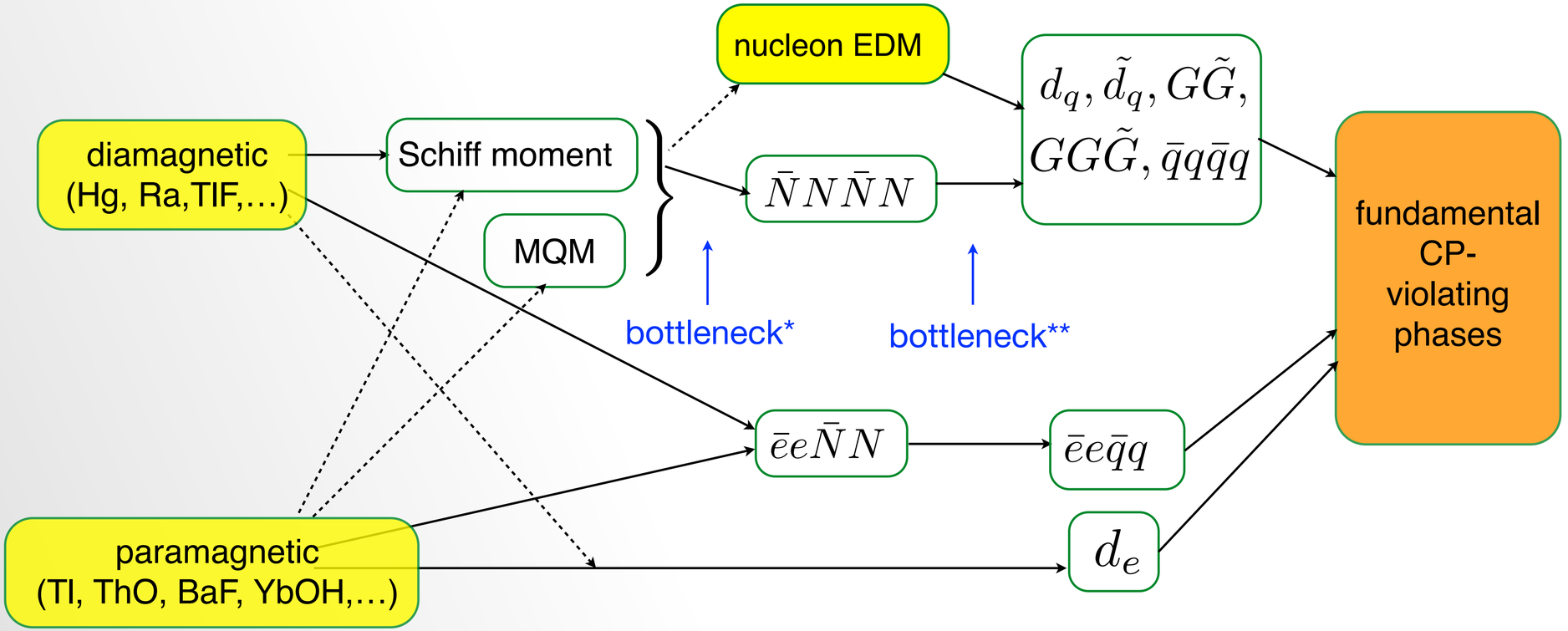
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* uncertainties up to 500% ** uncertainties ~100%

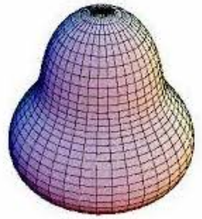
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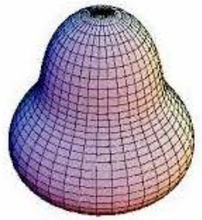
- Nuclear mechanism — octuple deformed nuclei, e.g., radium, radon

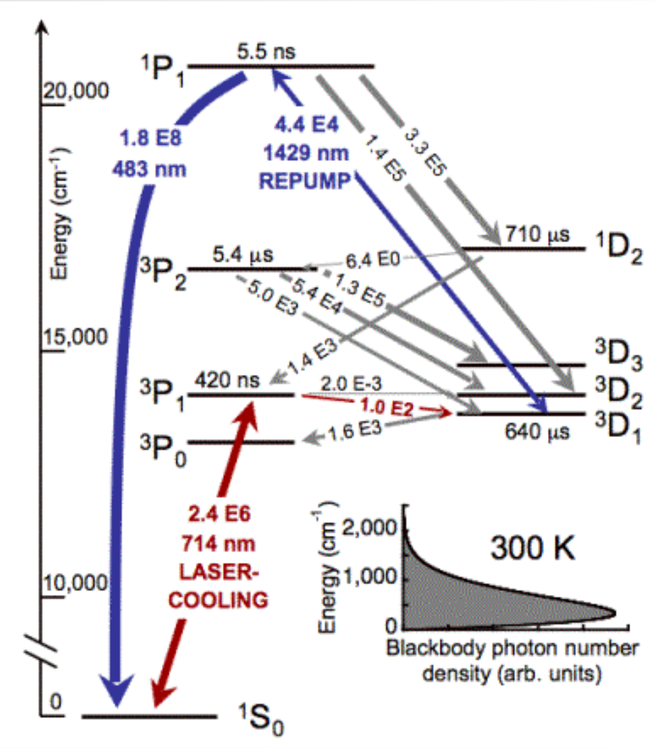


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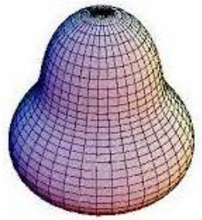
Argonne National Lab

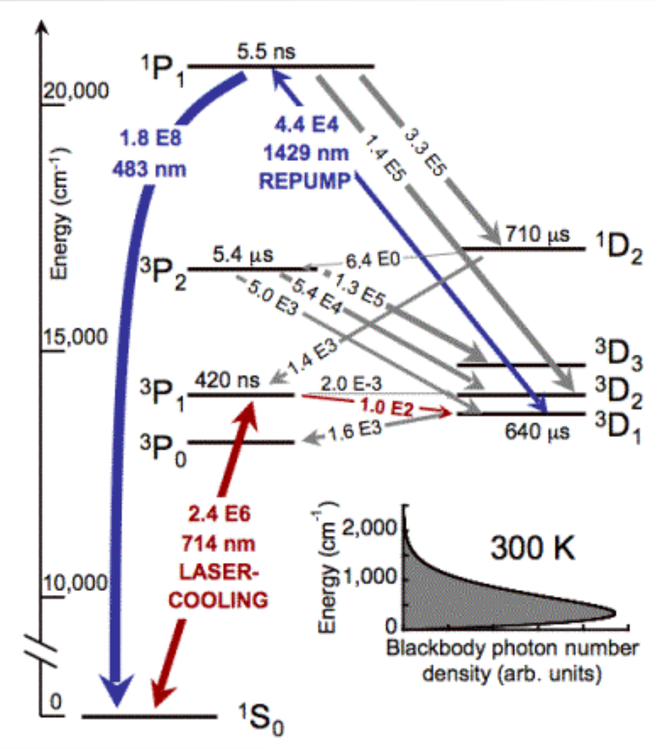
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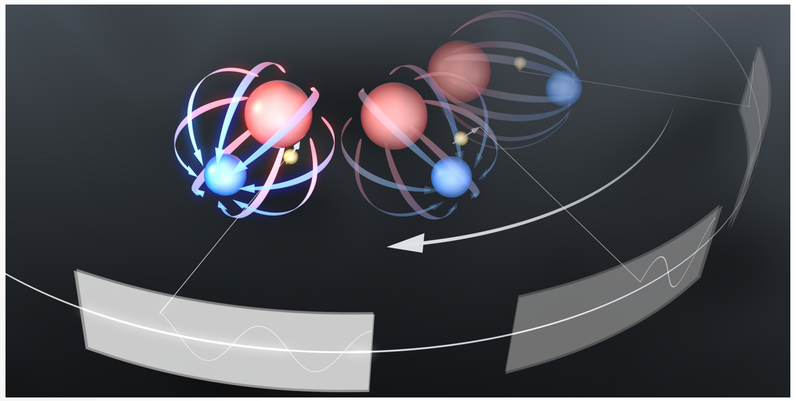




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- Electronic mechanism — close atomic levels of opposite parity, e.g., rare earth atoms, metastable state of radium,…

- Molecules — huge intramolecular electric field

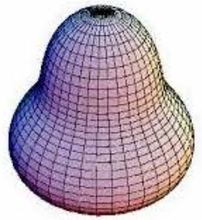


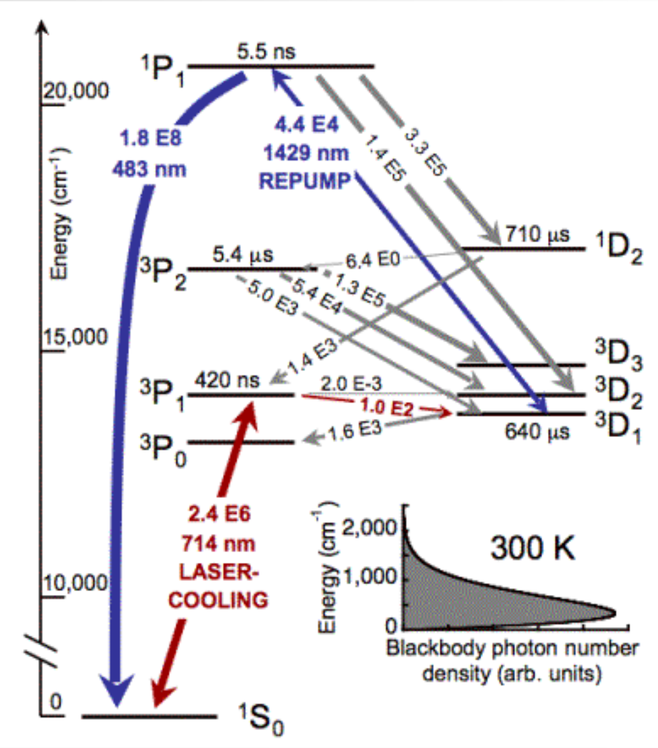
Eric Cornell group

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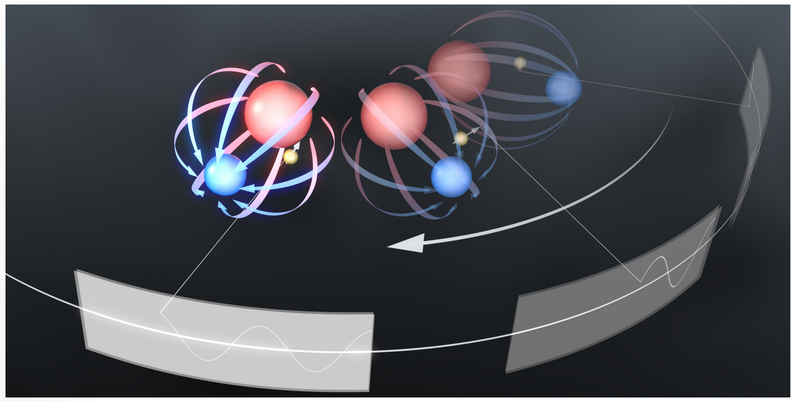




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Eric Cornell group

- Other — solids, oscillating electric field, …

Best limits and recent measurements

	$\lvert d_{atom}\rvert$, 95% c.l.	Constraints, 95% c.l.	Group
Paramagnetic			
^{205}Tl	$1.1 \times 10^{-24}\ e$ cm	$\lvert d_e\rvert < 1.9 \times 10^{-27}\ e$ cm	Berkeley, 2002
ThO		$\lvert d_e\rvert < 1.2 \times 10^{-29}\ e$ cm	ACME, 2018
HfF^+		$\lvert d_e\rvert < 4.1 \times 10^{-30}\ e$ cm (90% c.l)	JILA, 2023
Diamagnetic			
^{199}Hg	$7.4 \times 10^{-30}\ e$ cm	$\lvert d_n\rvert < 1.6 \times 10^{-26}\ e$ cm $\lvert\theta_{QCD}\rvert < 1.5 \times 10^{-10}$	Seattle, 2016
^{225}Ra	$1.4 \times 10^{-23}\ e$ cm		Argonne, 2015
^{129}Xe	$1.5 \times 10^{-27}\ e$ cm		Juelich, 2019
^{129}Xe	$4.8 \times 10^{-27}\ e$ cm		HeXeEDM, 2019
n		$\lvert d_n\rvert < 3.6 \times 10^{-26}\ e$ cm	Grenoble, 2015

Other ongoing experiments

Paramagnetic: YbF (Imperial), ThF (JILA), Cs (Penn State), Fr (Tokyo), BaF (Groningen) , YbOH (Caltech)

Diamagnetic: ^{223}Rn (TRIUMF), ^{129}Xe (Tokyo), TlF (CeNTREX)

+ many more, particularly for polyatomic molecules!

Resources:

Ginges and Flambaum, *Violations of fundamental symmetries and tests of unification theories of elementary particles*, Phys. Rep. 397, 63 (2004)

Safronova *et al*., *Search for new physics with atoms and molecules*, Rev. Mod. Phys. **90**, 025008 (2018)

Khriplovich, Parity nonconservation in atomic phenomena, Phys. Rev. C. **60,** 015501 (Gordon and Breach, Philadelphia, 1991)

Chupp, Fierlinger, Ramsey-Musolf, Singh, Rev. Mod. Phys. **91**, 015001 (2019).

After these lectures, you should:

- understand the basics of how weak interaction effects manifest in atoms, including in
  - atomic parity violation
  - time-reversal-violating electric dipole moments
- have an idea of the status and challenges of theory and experiment in this field